MIL-STD-242H(NAVY)-PART 1
25 March 1985
SUPERSEDING
MIL-STD-242G(NAVY)-PART 1

MILITARY STANDARD

ELECTRONIC EQUIPMENT PARTS
SELECTED STANDARDS

SYNCHROS, BLOWERS, AND ACOUSTICAL PARTS

NO DELIVERABLE DATA
REQUIRED BY THIS DOCUMENT

FSC 59GP

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ELECTRONIC EQUIPMENT PARTS, SELECTED STANDARDS, SYNCHROS, BLOWERS, AND ACOUSTICAL PARTS

MIL-STD-242H(NAVY), PART 1

1. THIS MILITARY STANDARD IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

2. RECOMMENDED CORRECTIONS, ADDITIONS OR DELETIONS SHOULD BE ADDRESSED TO: COMMANDER, NAVAL ELECTRONIC SYSTEMS COMMAND, ATTN: ELEX 8111, DEPARTMENT OF THE NAVY, WASHINGTON, D.C. 20360.

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FORWARD

THIS MILITARY STANDARD PART PROVIDES EQUIPMENT DESIGNERS AND MANUFACTURERS WITH A LIST OF SYNCHROS, BLOWERS, AND ACOUSTICAL PARTS HAVING QUALITY LEVELS MOST ACCEPTABLE IN THE DESIGN AND CONSTRUCTION OF MILITARY SYSTEMS AND EQUIPMENTS. IT ALSO WILL AID TO CONTROL AND MINIMIZE THE LIFE CYCLE COST, IMPROVE RELIABILITY OF EQUIPMENT AND MINIMIZE LOGISTIC SUPPORT. THE CRITERIA USED IN SELECTING PARTS ARE AS FOLLOWS: -

1. APPLICATION NEED: PARTS MUST SATISFY THE WIDEST RANGE OF DESIGN REQUIRING RELIABLE PARTS.

2. TECHNOLOGICAL MATURITY: THE DESIGN OF THE PART MUST BE FINAL AND MUST USE PROVEN MATERIALS AND TECHNOLOGIES. IT MUST HAVE BEEN IN PRODUCTION FOR A PERIOD SUFFICIENT TO ENSURE THAT THE DESIGN AND PROCESS PARAMETERS HAVE BEEN IDENTIFIED AND ADEQUATE QUALITY CONTROLS HAVE BEEN DEVELOPED.

3. AVAILABILITY: THE PART MUST BE IN PRODUCTION BY AT LEAST ONE MANUFACTURER WHOSE PREVIOUS PERFORMANCE INDICATES ABILITY TO QUALIFY TO SPECIFICATIONS OF THIS STANDARD. THERE MUST ALSO BE REASONABLE EXPECTATION THAT THE PART WILL NOT BE OBSOLESCENT FOR AT LEAST SEVEN YEARS. MICROCIRCUITS ARE EXCLUDED FROM THIS REQUIREMENT BECAUSE OF RAPID TECHNOLOGICAL CHANGES.

4. TEST OR USAGE HISTORY: SUFFICIENT TEST OR USAGE DATA TO PREDICT PART RELIABILITY MUST BE AVAILABLE.

IN THE EVENT OF CONFLICT BETWEEN THE TECHNICAL DESCRIPTION OF STANDARD PARTS IN THIS STANDARD AND THE APPLICABLE SPECIFICATION, THE SPECIFICATION SHALL GOVERN.

TECHNICAL INFORMATION INCLUDED IN THIS STANDARD WAS OBTAINED FROM MILITARY SPECIFICATIONS AND STANDARDS: THE DRAWINGS SHOWN MAY NOT NECESSARILY BE EXACTLY TO SCALE. SOME DRAWINGS MAY SHOW EXAGGERATED FEATURES TO DEPICT CLARITY. NO WARRANTY IS MADE OF DATA ACCURACY, OR THAT INCLUSION OF THESE PARTS WILL ASSURE EQUIPMENT/SYSTEMS WILL MEET PERFORMANCE REQUIREMENTS OF ANY CONTRACTS. THE CONTRACTOR IS RESPONSIBLE FOR CONDUCTING NECESSARY TESTS AND INSPECTIONS OF SELECTED PARTS TO ASSURE THAT CONTRACT REQUIREMENTS ARE MET.

REVIEW ACTIVITIES:
NAVY - AS, SH, OS

USER ACTIVITY:
NAVY - MC

PREPARING ACTIVITY:
NAVY - EC

(PROJECT NUMBER 59GP-N051)

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SYNCHROS, BLOWERS, AND ACOUSTICAL PARTS

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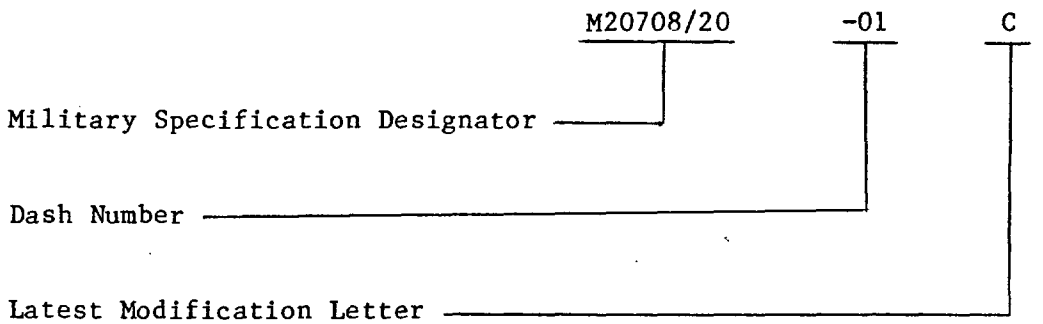
SUBJECT/SECTION			PAGE NUMBER
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	1 0	MIL-S-20708, 400 Hz	120.1 - 120.15
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SYNCHROS

SCOPE: This section covers SYNCHROS, 60 and 400 Hz. These SYNCHROS shall be used in conjunction with MIL-S-81963.

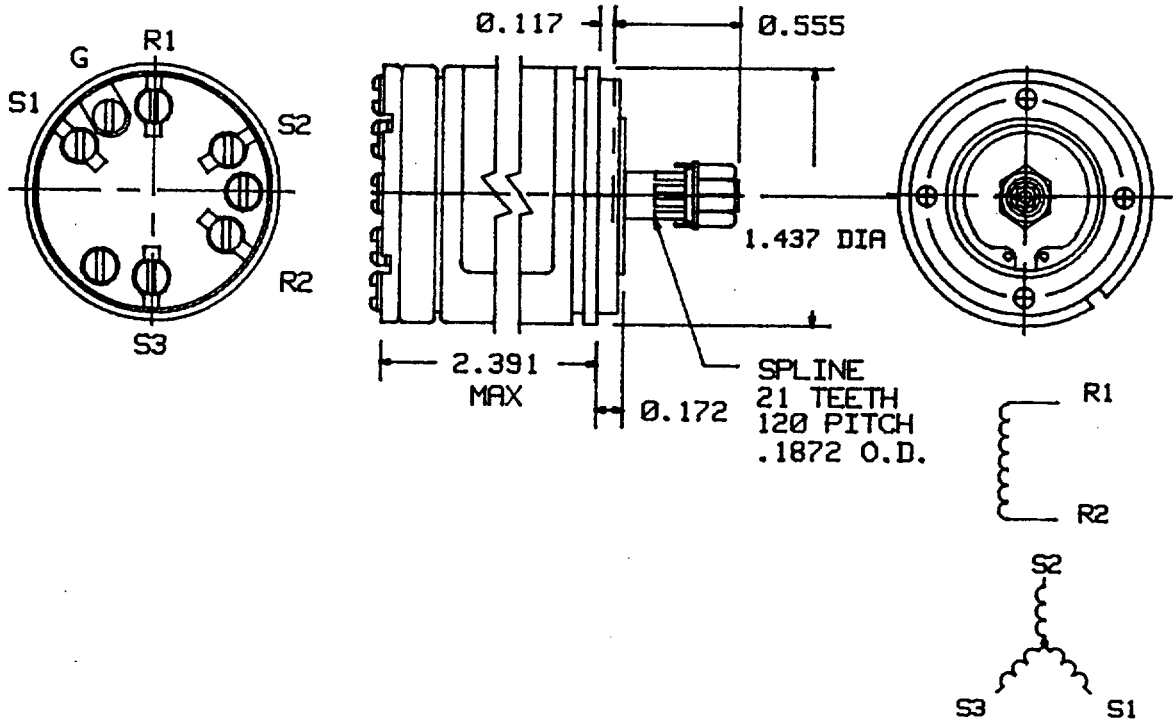
PART NUMBER EXAMPLE: M20708/20-01C



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SYNCHRO, CONTROL TRANSMITTER, TYPE 15CX6C

MIL-S-20708/20



CIRCUIT DIAGRAM

CHARACTERISTICS:

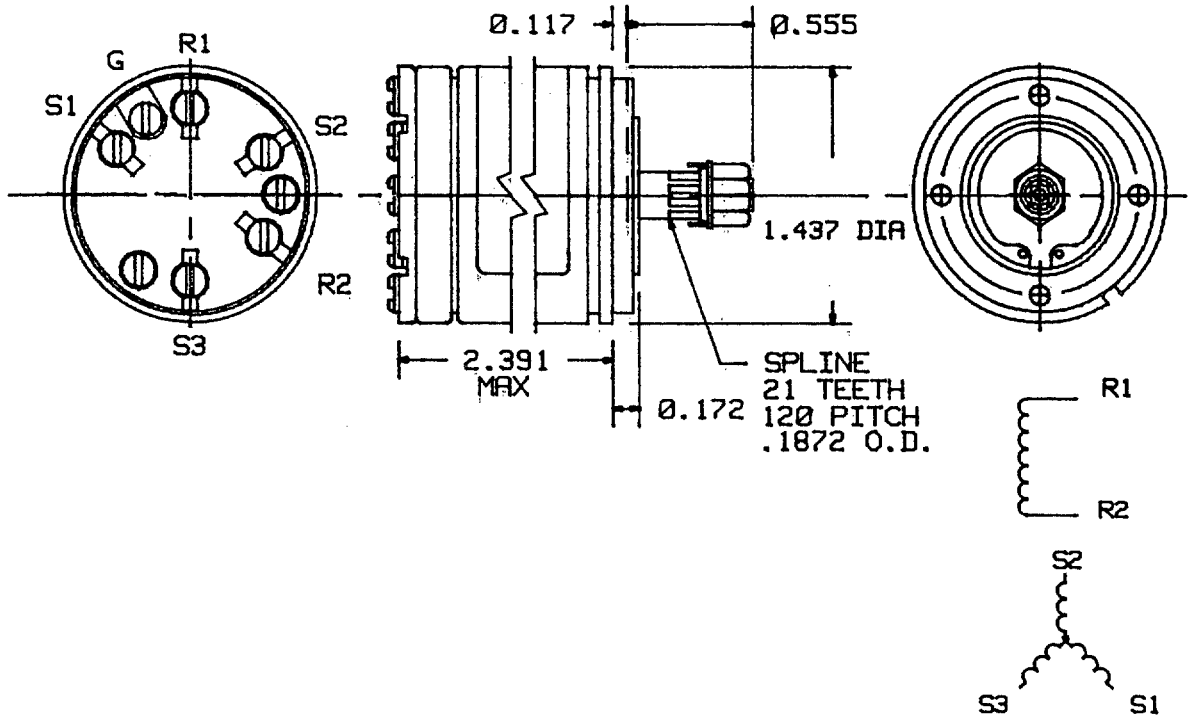
PART NUMBER: M20708/20-01C

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 22.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.36 WATT
IMPEDANCE:
ZRO: 5,800 OHMS
ZSS: 750 OHMS
IMPEDANCE ANGLE:
ZRO: 81.5 DEGREES
ZSS: 13.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 7.0 DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 90.0 MILLIVOLTS
FUNDAMENTAL (MAX): 45.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.05 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 15CT6D

MIL-S-20708/21



CIRCUIT DIAGRAM

CHARACTERISTICS:

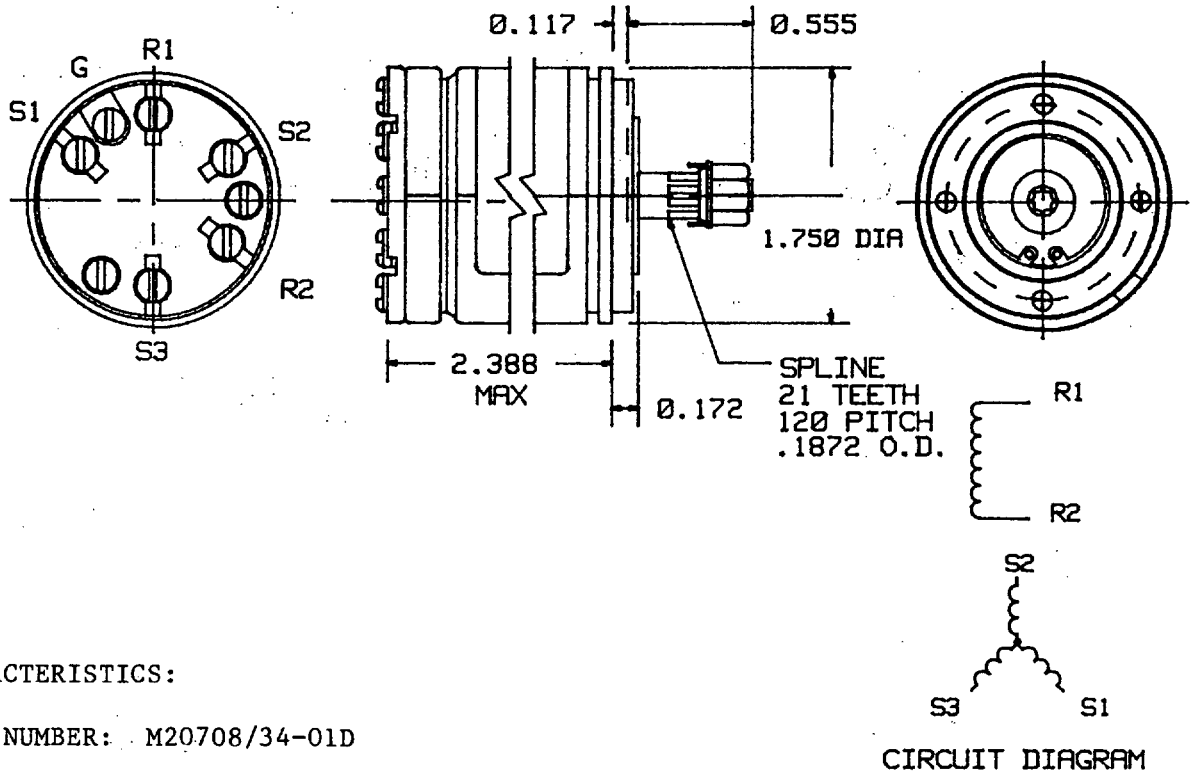
PART NUMBER: M20708/21-01D

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 13 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.21 WATT
IMPEDANCE:
ZRS: 1,475 OHMS
ZSO: 6,750 OHMS
IMPEDANCE ANGLE:
ZRO: 15.5 DEGREES
ZSS: 80 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 9.0 DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 65.0 MILLIVOLTS
FUNDAMENTAL (MAX): 45.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.05 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 18CT6D

MIL-S-20708/34



CHARACTERISTICS:

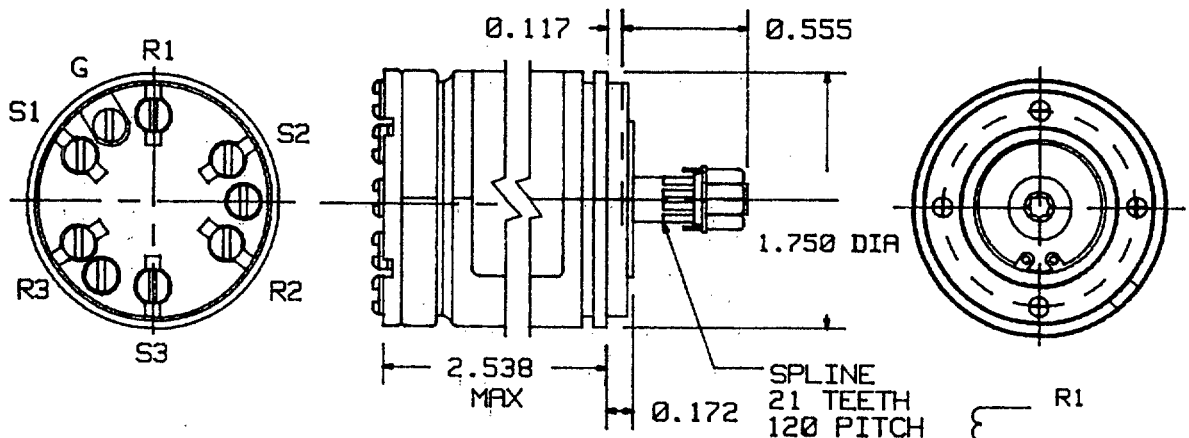
PART NUMBER: M20708/34-01D

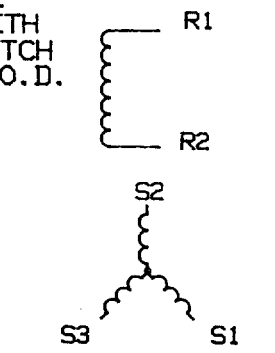
FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 17.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.36 WATTS
IMPEDANCE:
ZSO: 5,200 OHMS
ZRS: 1,825 OHMS
IMPEDANCE ANGLE:
ZSO: 73.0 DEGREES
ZRS: 16.3 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 16.0 DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 45.0 MILLIVOLTS
FUNDAMENTAL (MAX): 25.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 18CDX6D

MIL-S-20708/36





CIRCUIT DIAGRAM

CHARACTERISTICS:

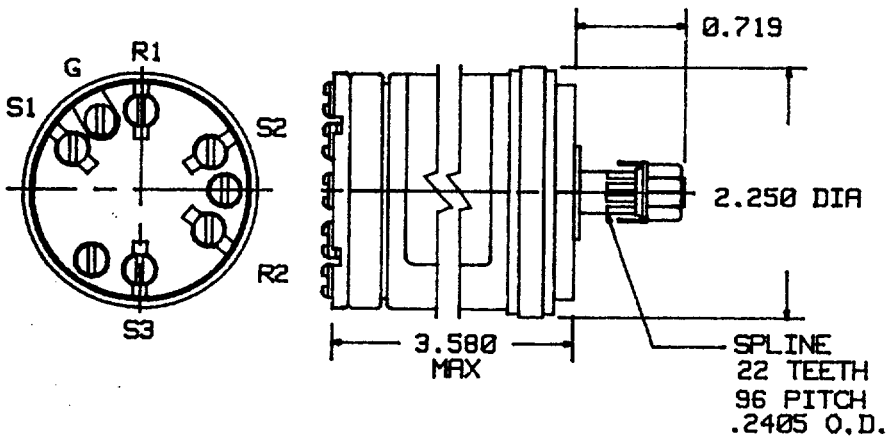
PART NUMBER: M20708/36C-1

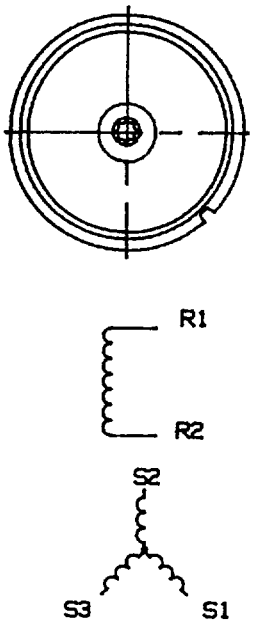
FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 52.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 1.10 WATTS
IMPEDANCE:
ZSO: 1,687 OHMS
ZRS: 1,070 OHMS
IMPEDANCE ANGLE:
ZSO: 72.0 DEGREES
ZRS: 15.0 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 15.5 DEGREES
ELECTRICAL ERROR (MAX): 7.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 100 MILLIVOLTS
FUNDAMENTAL (MAX): 60.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL TRANSMITTER, TYPE 23CX6D

MIL-S-20708/52





CIRCUIT DIAGRAM

CHARACTERISTICS:

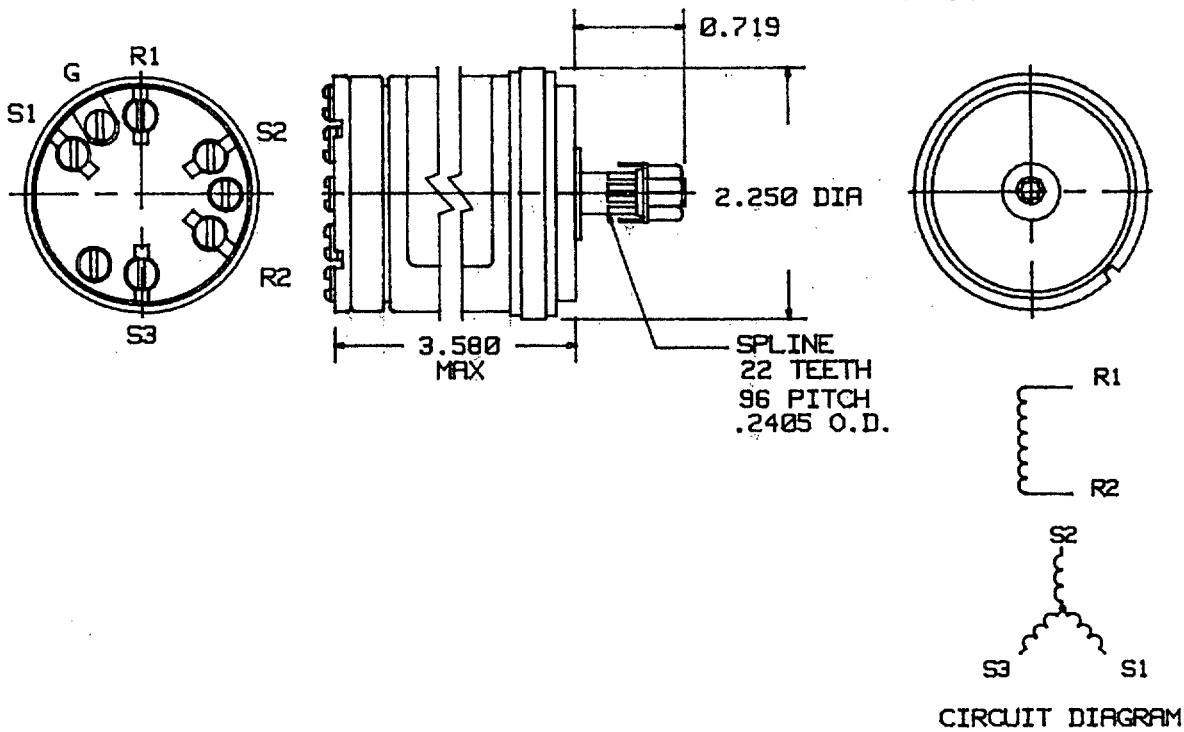
PART NUMBER: M20708/52-01D

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 80 MILLIAMPS
PRIMARY POWER (NOMINAL): 1.40 WATTS
IMPEDANCE:
ZRO: 1,575 OHMS
ZSS: 232.5 OHMS
IMPEDANCE ANGLE:
ZRO: 81.0 DEGREES
ZSS: 12.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 6.5 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 30.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 23CT6D

MIL-S-20708/53



CIRCUIT DIAGRAM

CHARACTERISTICS:

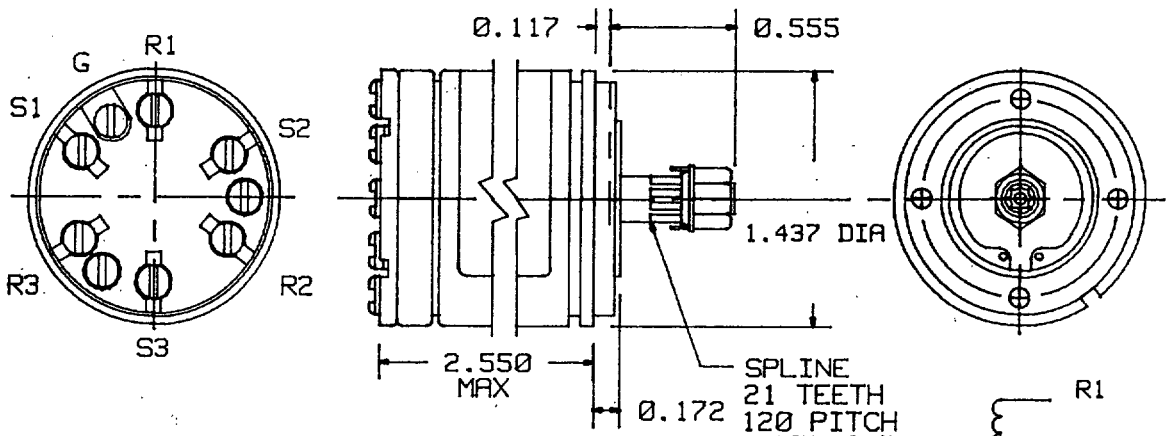
PART NUMBER: M20708/53-01D

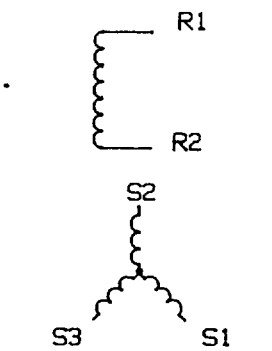
FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 18.5 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.40 WATTS
IMPEDANCE:
ZSO (MIN-MAX): 4,650 OHMS
ZRS (MIN-MAX): 1,500 OHMS
IMPEDANCE ANGLE:
ZSO (MIN-MAX): 73 DEGREES
ZRS (MIN-MAX): 17 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 14 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 45.0 MILLIVOLTS
FUNDAMENTAL (MAX): 30.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER TYPE 15CDX6C

MIL-S-20708/22





CIRCUIT DIAGRAM

CHARACTERISTICS:

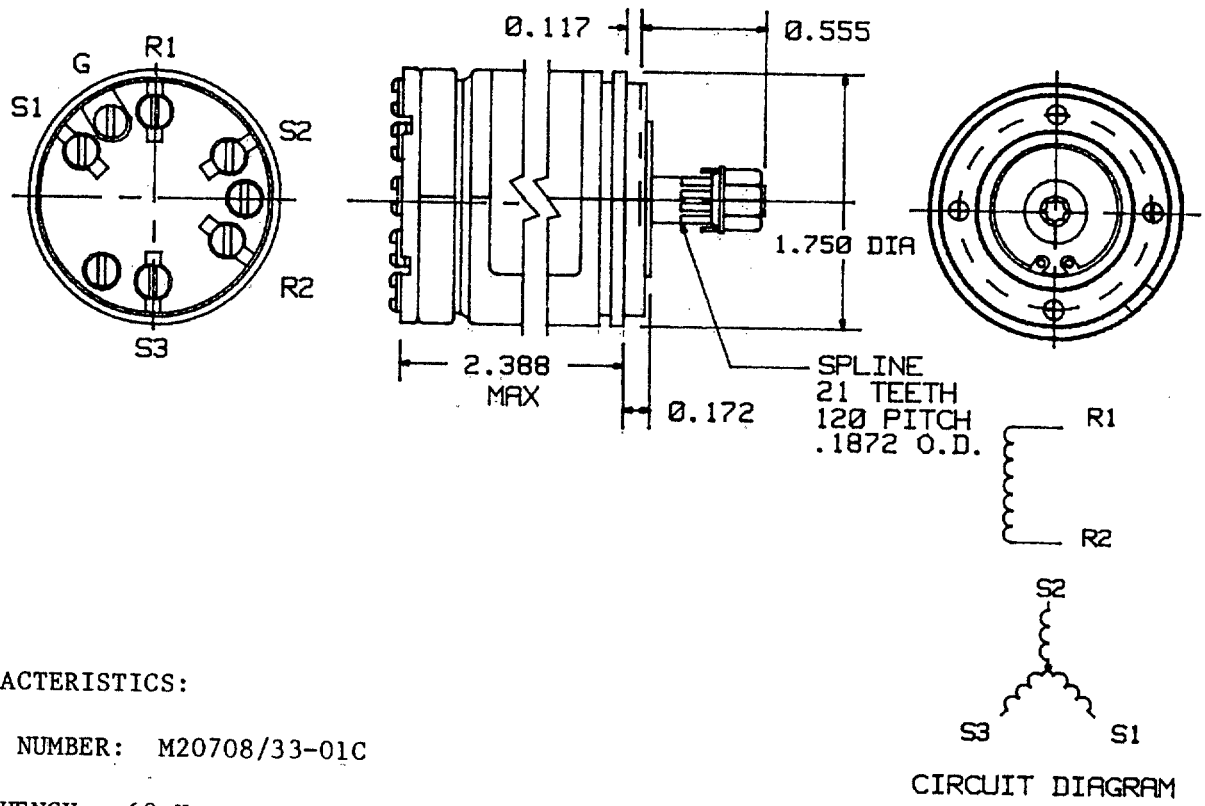
PART NUMBER: M20708/22-01C

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 38 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.50 WATT
IMPEDANCE:
ZSO: 2,325 OHMS
ZRS: 1,050 OHMS
IMPEDANCE ANGLE:
ZSO: 80 DEGREES
ZRS: 14 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 9.0 DEGREES
ELECTRICAL ERROR (MAX): 7.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 100 MILLIVOLTS
FUNDAMENTAL (MAX): 60.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.07 OUNCE-INCH

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SYNCHRO, CONTROL TRANSMITTER, TYPE 18CX6C

MIL-S-20708/33



CIRCUIT DIAGRAM

CHARACTERISTICS:

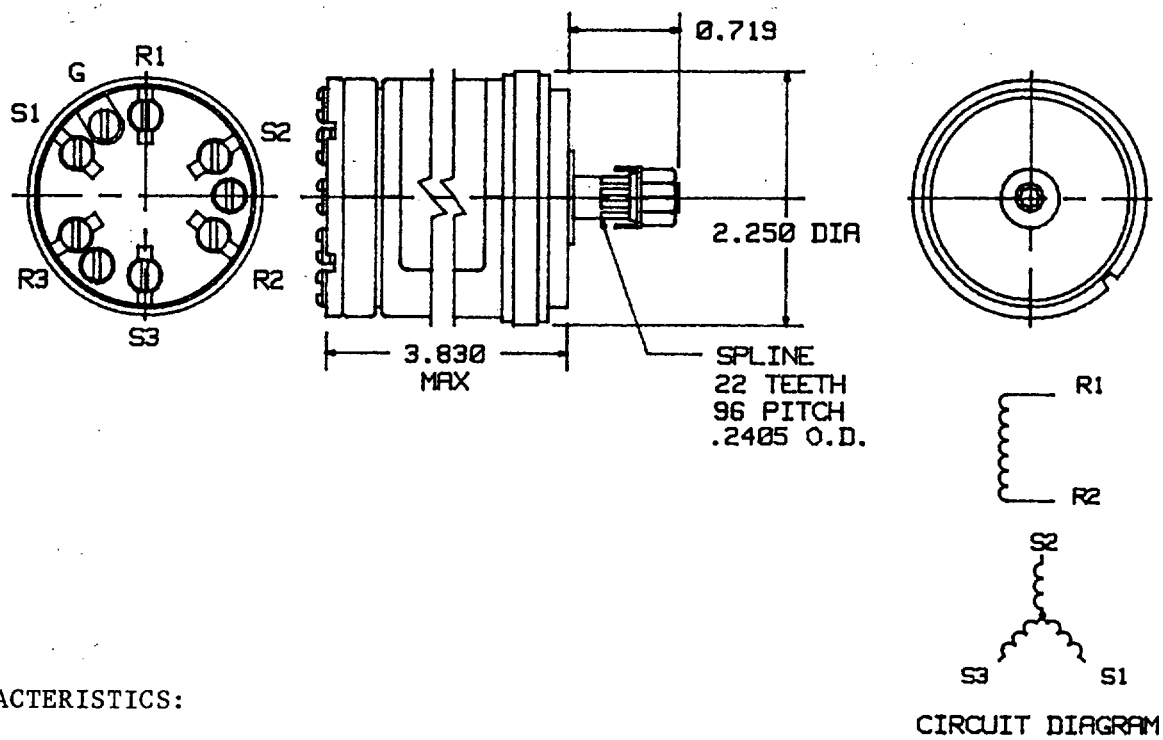
PART NUMBER: M20708/33-01C

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 40.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.95 WATT
IMPEDANCE:
    ZRO: 3,125 OHMS
    ZSS: 735 OHMS
IMPEDANCE ANGLE:
    ZRO: 78.0 DEGREES
    ZSS: 11.0 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 11.0 DEGREES
ELECTRICAL ERROR (MAX): 8.0 MINUTES
NULL VOLTAGE:
    TOTAL (MAX): 85.0 MILLIVOLTS
    FUNDAMENTAL (MAX): 30.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 23CDX6C

MIL-S-20708/54



CIRCUIT DIAGRAM

CHARACTERISTICS:

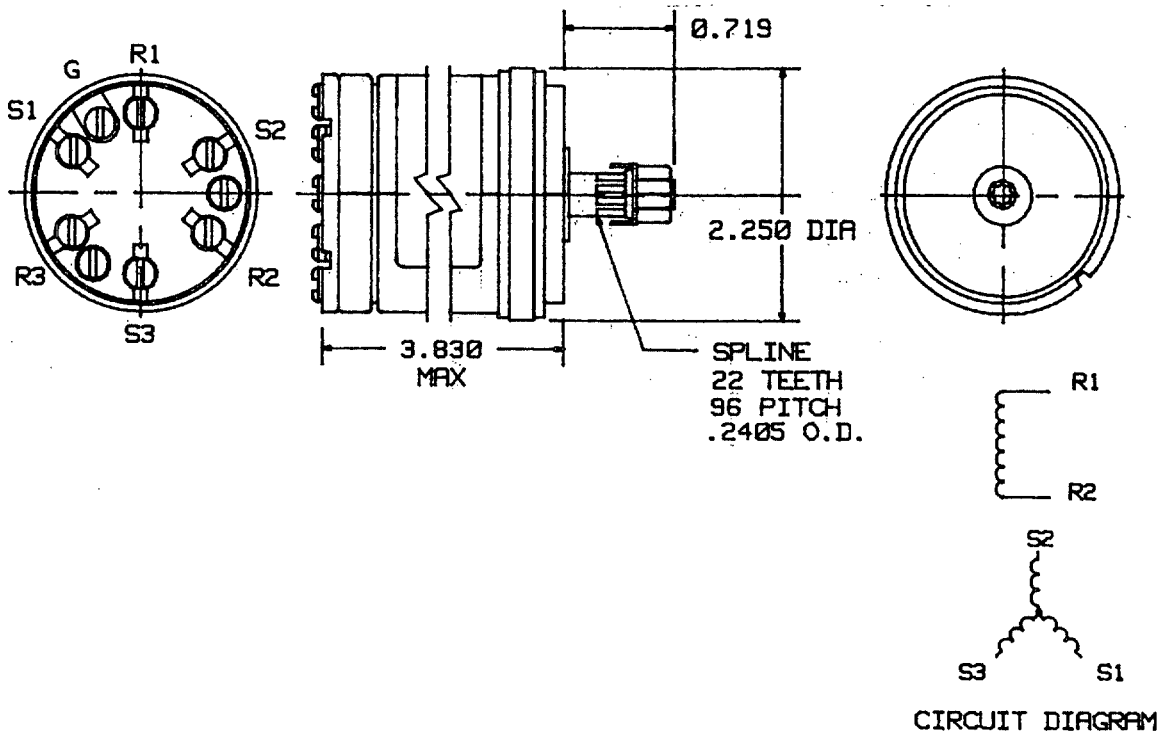
PART NUMBER: M20708/54-01C

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 90.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 1.40 WATTS
IMPEDANCE:
ZSO: 983.5 OHMS
ZRS: 449.0 OHMS
IMPEDANCE ANGLE:
ZSO: 77 DEGEES
ZRS: 19 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 11.0 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 8.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 65 MILLIVOLTS
FUNDAMENTAL (MAX): 40.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, TORQUE DIFFERENTIAL TRANSMITTER, TYPE 23TDX6C

MIL-S-20708/55



CHARACTERISTICS:

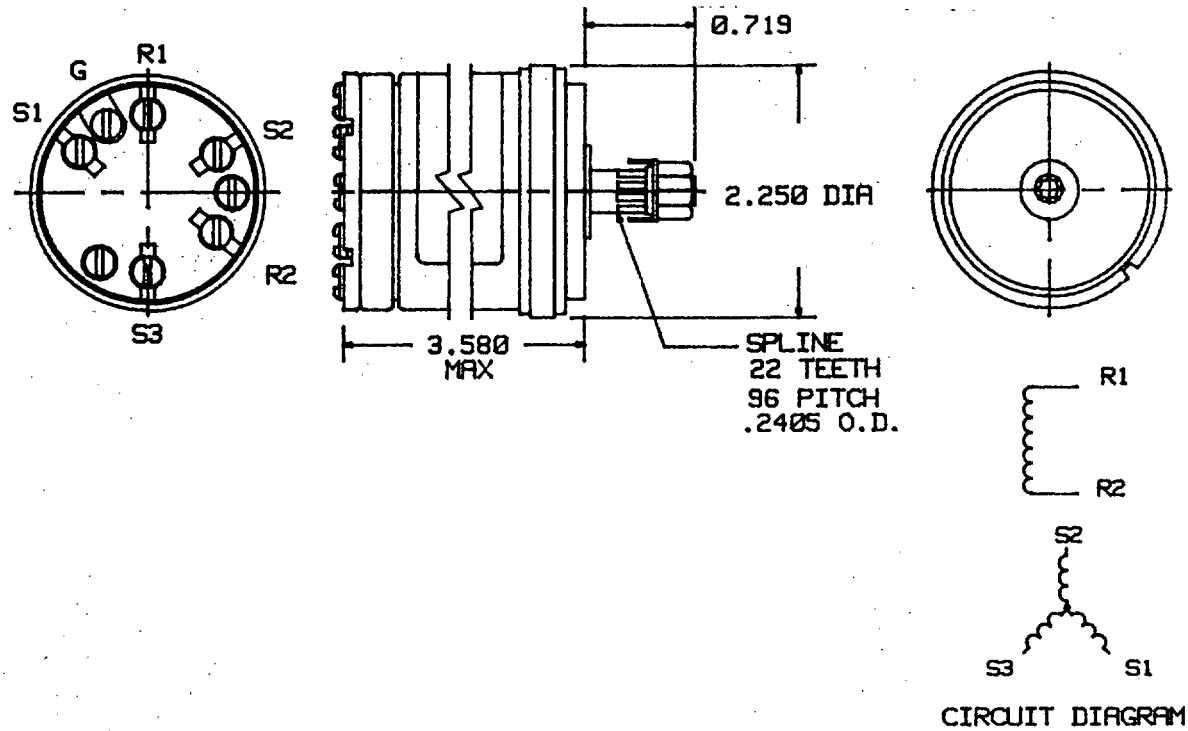
PART NUMBER: M20708/55-01C

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT (MAX): 215 MILLIAMPS
PRIMARY POWER (MAX): 4.8 WATTS
IMPEDANCE:
ZSO: 415 OHMS
ZRS: 197 OHMS
IMPEDANCE ANGLE:
ZSO: 76.0 DEGREES
ZRS: 17.5 DEGREES
TRANSFORMATION RATIO: 1,154
PHASE SHIFT: 11.50 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 6.0 MINUTES
TORQUE GRADIENT (MIN): 0.03 OUNCE-INCH/DEGREE
FRICTION TORQUE (MAX): 0.2 OUNCE/INCH

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 23TRX6B

MIL-S-20708/56



CIRCUIT DIAGRAM

CHARACTERISTICS:

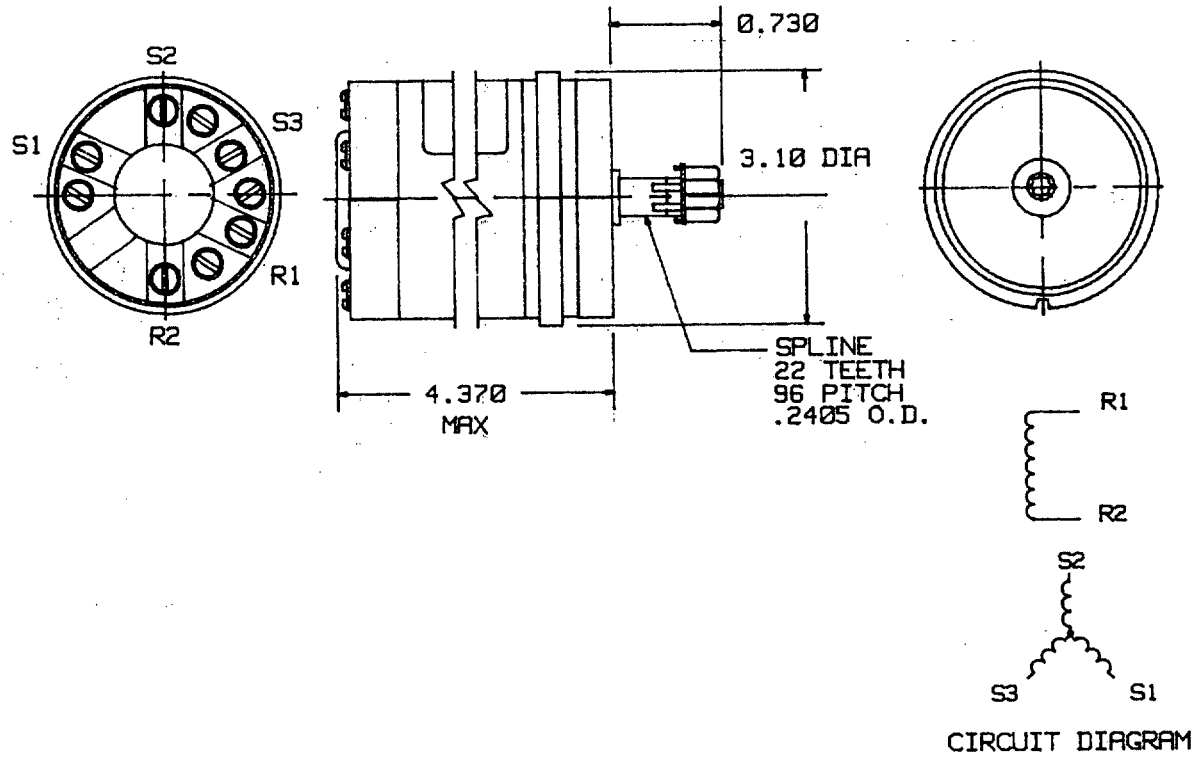
PART NUMBER: M20708/56-01B

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT (MAX): 210 MILLIAMPS
PRIMARY POWER: 5.60 WATTS
IMPEDANCE:
ZSS: 127.5 OHMS
IMPEDANCE ANGLE:
ZSS: 13.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 11.0 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 8 MINUTES
RECEIVER ERROR (MAX): 45 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 160.0 MILLIVOLTS
FUNDAMENTAL (MAX): 60.0 MILLIVOLTS
TORQUE GRADIENT (MIN): 0.12 OUNCE-INCH/DEGREE

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 31TRX6A

MIL-S-20708/66



CIRCUIT DIAGRAM

CHARACTERISTICS:

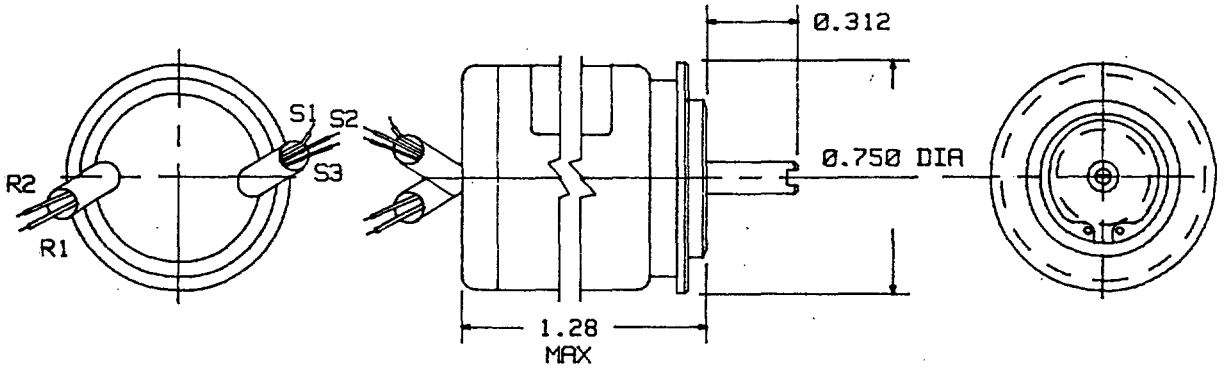
PART NUMBER: M20708/66-01A

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT (MAX): 440 MILLIAMPS
PRIMARY POWER: 6.60 WATTS
IMPEDANCE:
ZSS: 28.5 OHMS
IMPEDANCE ANGLE:
ZSS: 20.0 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 6.5 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 10 MINUTES
RECEIVER ERROR (MAX): 36 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 170.0 MILLIVOLTS
FUNDAMENTAL (MAX): 35.0 MILLIVOLTS
TORQUE GRADIENT (MIN): 0.4 OUNCE-INCH/DEGREE

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SYNCHRO, CONTROL TRANSMITTER, TYPE 08CX4C

MIL-S-20708/78



CHARACTERISTICS:

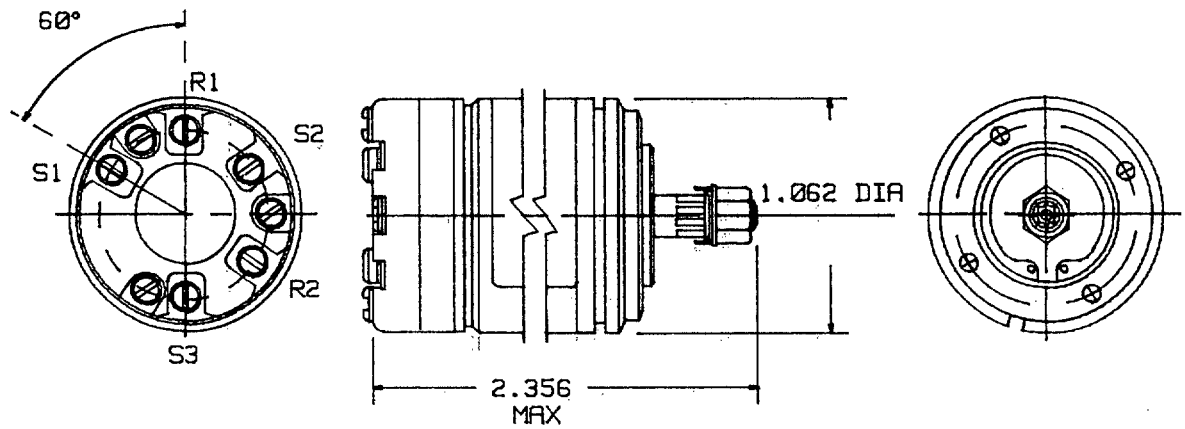
PART NUMBER: M20708/78-01C

FREQUENCY: 400 Hz
PRIMARY VOLTAGE: 26 VOLTS
PRIMARY CURRENT (MAX): 153.0 MILLIAMPS
PRIMARY POWER: 0.70 WATTS NOMINAL
IMPEDANCE:
ZRO: 170.0-215.0 OHMS / 77.5-80.5 DEGREES
ZSS: 11.0-15.0 OHMS / 9.0-15.0 DEGREES
TRANSFORMATION RATIO: 0.454
PHASE SHIFT: 8.5 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 7.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 30.0 MILLIVOLTS
FUNDAMENTAL (MAX): 20.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.04 OUNCE-INCH

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SYNCHRO, CONTROL TRANSMITTER, TYPE 11CX4E

MIL-S-20708/2



CHARACTERISTICS:

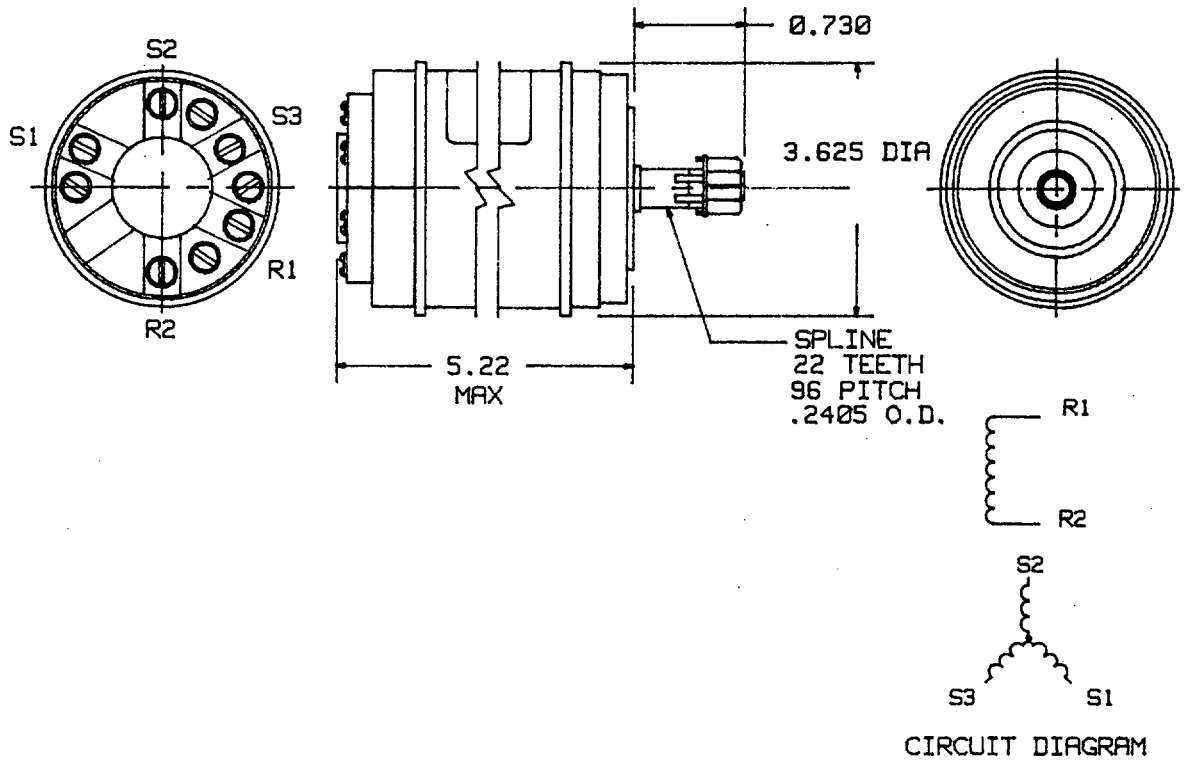
PART NUMBER: M20708/2-01E

FREQUENCY: 400 Hz
PRIMARY VOLTAGE: 115 VOLTS
PRIMARY CURRENT (MAX): 31 MILLIAMPS
PRIMARY POWER: 0.49 WATTS NOMINAL
IMPEDANCE:
ZRO: 3720-4700 OHMS / 80-84 DEGREES
ZSS: 410-500 OHMS / 17-26 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 4.5 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 7.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 75.0 MILLIVOLTS
FUNDAMENTAL (MAX): 45.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.07 OUNCE-INCH

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 37TRX6A

MIL-S-20708/74



CIRCUIT DIAGRAM

CHARACTERISTICS:

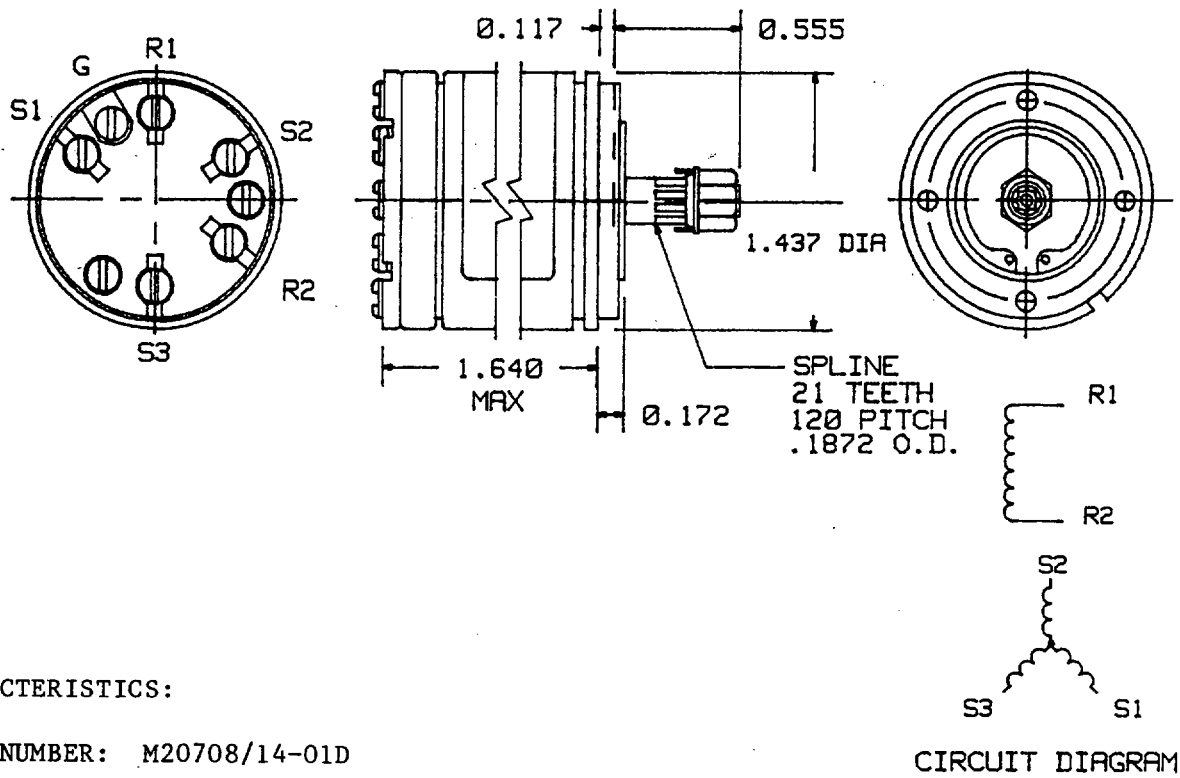
PART NUMBER: M20708/74-01A

FREQUENCY: 60 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT (MAX): 830 MILLIAMPS
PRIMARY POWER: 9.25 WATTS
IMPEDANCE:
ZSS: 8.5 OHMS
IMPEDANCE ANGLE:
ZSS: 35.0 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 3.5 (LEAD) DEGREES MAX
RECEIVER ERROR (MAX): 36 MINUTESS
NULL VOLTAGE:
TOTAL (MAX): 170.0 MILLIVOLTS
FUNDAMENTAL (MAX): 35.0 MILLIVOLTS
TORQUE GRADIENT (MIN): 0.90 OUNCE-INCH/DEGREE

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SYNCHRO, CONTROL TRANSMITTER, TYPE 15CX4D

MIL-S-20708/14



CIRCUIT DIAGRAM

CHARACTERISTICS:

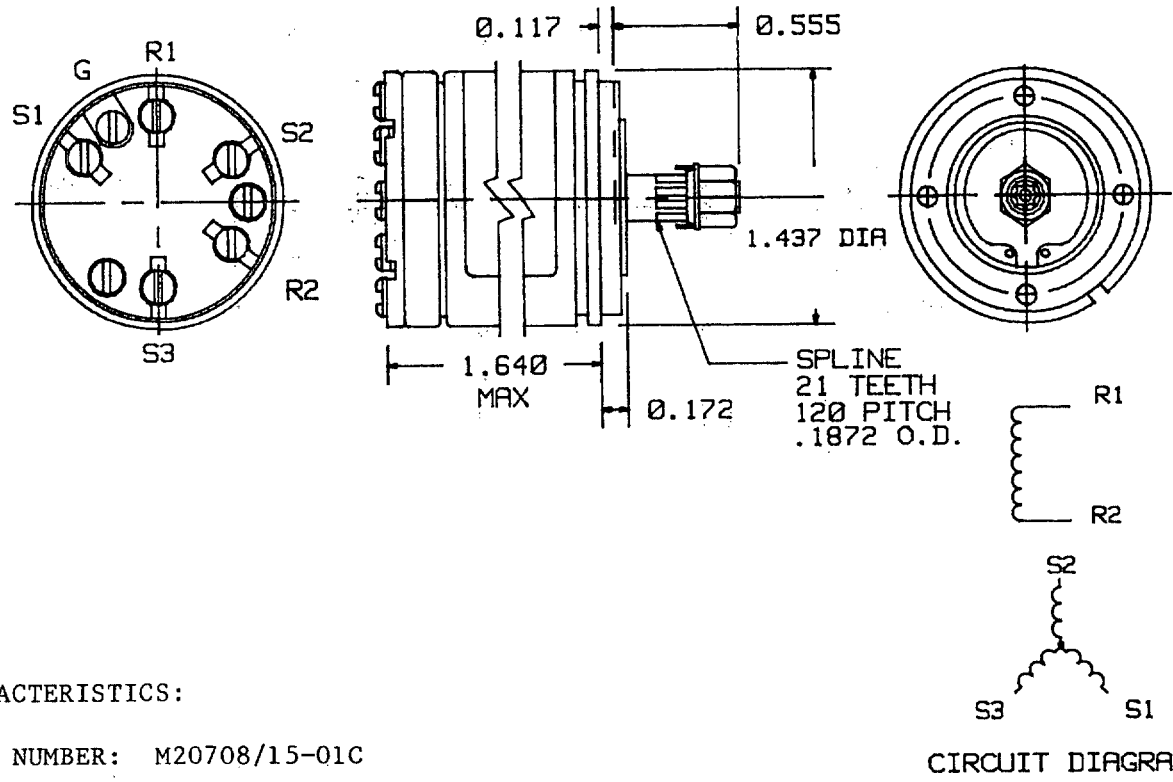
PART NUMBER: M20708/14-01D

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 85 MILLIAMPS
PRIMARY POWER (NOMINAL): 1.0 WATT
IMPEDANCE:
ZRO (MIN-MAX): 1,402-1,643
ZSS (MIN-MAX); 123-138 OHMS
IMPEDANCE ANGLE:
ZRO (MIN-MAX): 81.4-84.6 DEGREES
ZSS (MIN-MAX): 26.5-31.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 4.0 (LEAD) DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 32.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.05 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 15CT4C

MIL-S-20708/15



CIRCUIT DIAGRAM

CHARACTERISTICS:

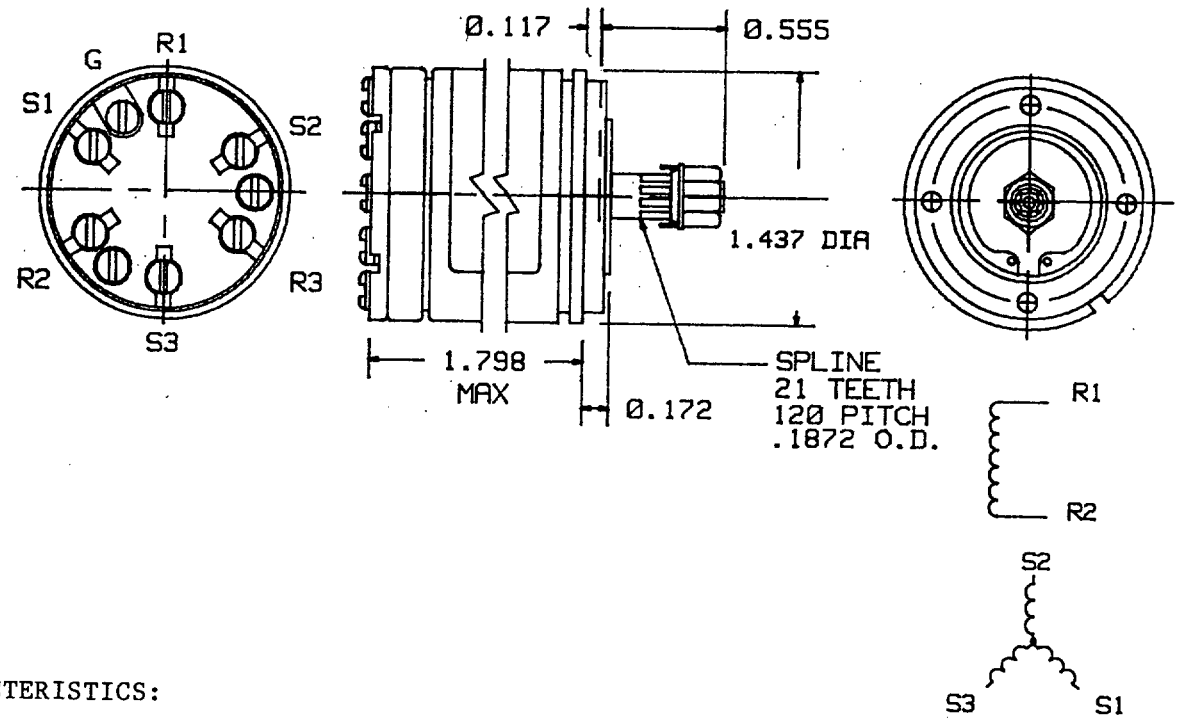
PART NUMBER: M20708/15-01C

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 10 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.14 WATT
IMPEDANCE:
ZSO (MIN-MAX): 7,800-9,512 OHMS
ZRS (MIN-MAX); 1,050-1,300 OHMS
IMPEDANCE ANGLE:
ZSO (MIN-MAX): 77.8-84.5 DEGREES
ZRS (MIN-MAX): 26.0-33.0 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 5.5 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 32.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.05 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 15CDX4D

MIL-S-20708/16



CIRCUIT DIAGRAM

CHARACTERISTICS:

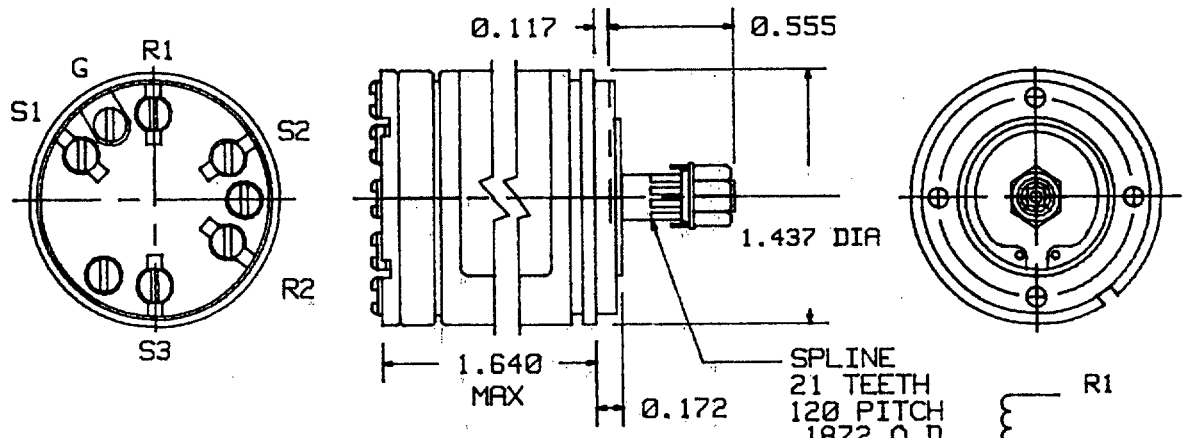
PART NUMBER: M20708/16-01D

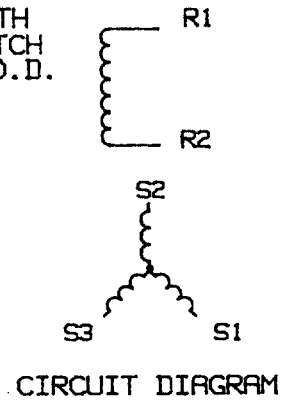
FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 90 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.9 WATT
IMPEDANCE:
ZSO: 1,010 OHMS
ZRS: 242.5 OHMS
IMPEDANCE ANGLE:
ZSO: 82 DEGREES
ZRS: 32 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 5.5 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 32.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.07 OUNCE-INCH

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 15TRX4A

MIL-S-20708/19





CIRCUIT DIAGRAM

CHARACTERISTICS:

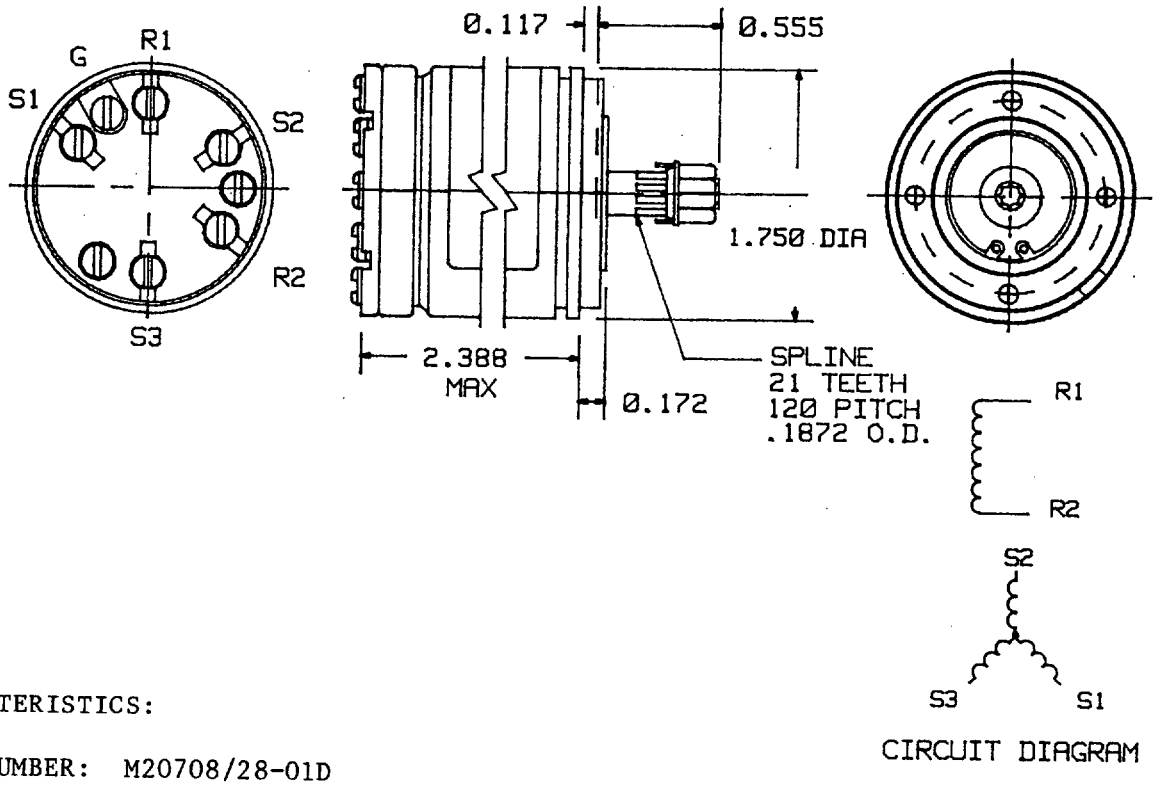
PART NUMBER: M20708/19-01A

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 190 MILLIAMPS
PRIMARY POWER (NOMINAL): 2.9 WATT
IMPEDANCE:
ZSS: 66 OHMS
IMPEDANCE ANGLE:
ZSS: 39.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 5.0 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 100.0 MILLIVOLTS
FUNDAMENTAL (MAX): 50.0 MILLIVOLTS
RECEIVER ERROR (MAX): 45 MINUTES
TORQUE GRADIENT (MIN): 0.03 OUNCE-INCH/DEGREE

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SYNCHRO, CONTROL TRANSMITTER, TYPE 18CX4D

MIL-S-20708/28



CIRCUIT DIAGRAM

CHARACTERISTICS:

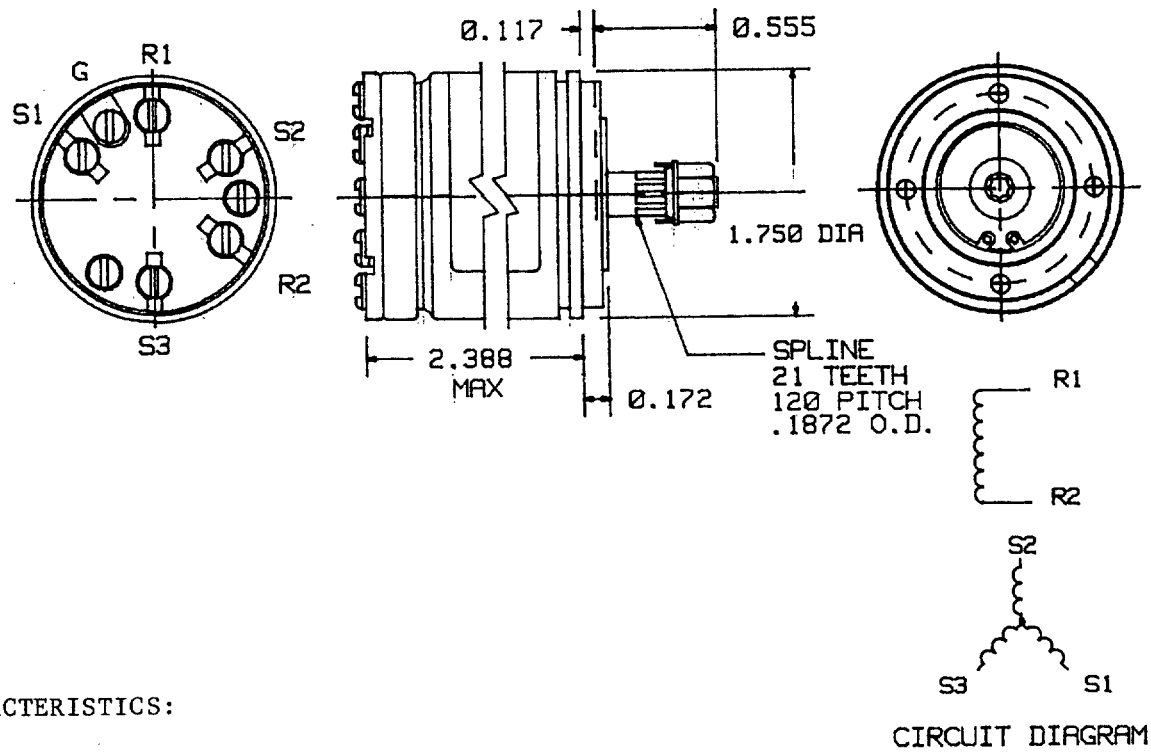
PART NUMBER: M20708/28-01D

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 110 MILLIAMPS MAX
PRIMARY POWER (NOMINAL): 0.9 WATT
IMPEDANCE:
ZRO: 1,222.5 OHMS
ZSS: 48.5 OHMS
IMPEDANCE ANGLE:
ZRO: 85.7 DEGREES
ZSS: 41.0 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 1.5 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 40.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 18CT4C

MIL-S-20708/29



CIRCUIT DIAGRAM

CHARACTERISTICS:

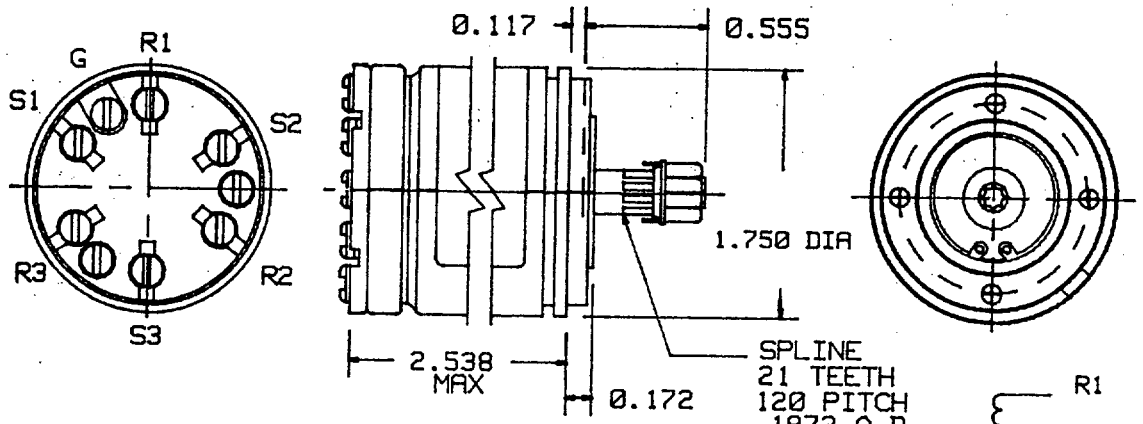
PART NUMBER: M20708/29-01C

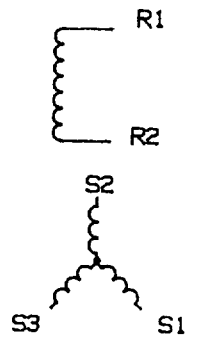
FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 7.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.05 WATT
IMPEDANCE:
ZSO: 13,400 OHMS
ZRS: 1,085 OHMS
IMPEDANCE ANGLE:
ZSO: 83.3 DEGREES
ZRS: 47.0 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 2.0 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 30.0 MILLIVOLTS
FUNDAMENTAL (MAX): 20.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 18CDX4C

MIL-S-20708/30





CIRCUIT DIAGRAM

CHARACTERISTICS:

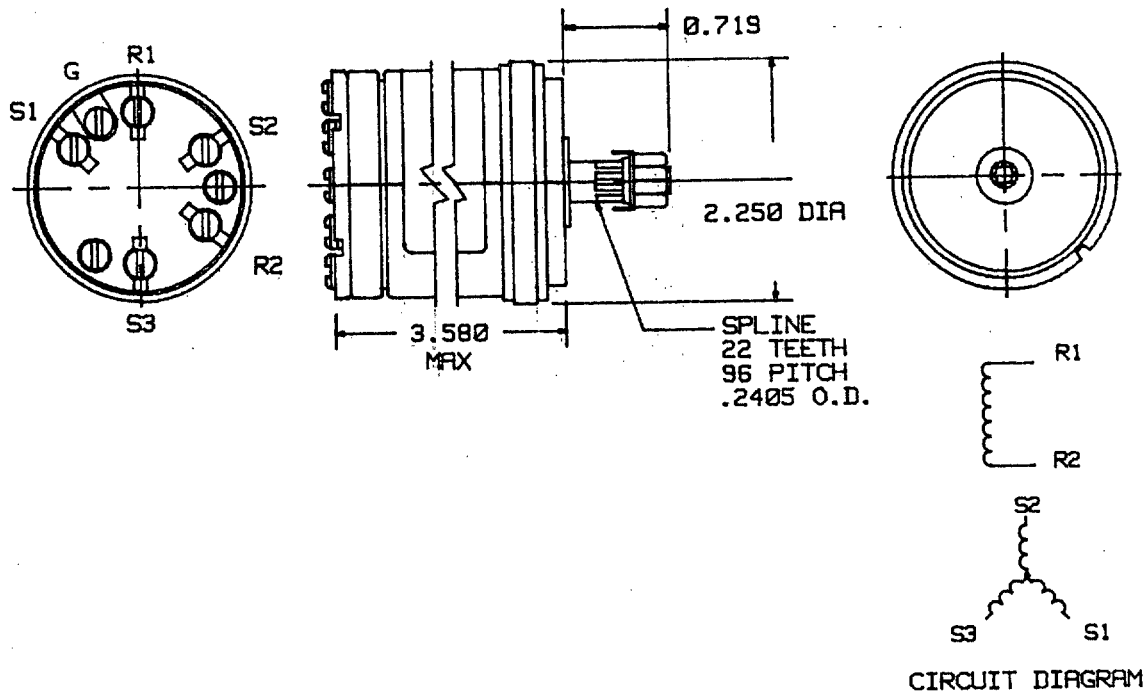
PART NUMBER: M20708/30-01C

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 128.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.95 WATT
IMPEDANCE:
ZSO: 697.5 OHMS
ZRS: 93.0 OHMS
IMPEDANCE ANGLE:
ZSO: 84.8 DEGREES
ZRS: 47.5 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 2.0 DEGREES
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 75.0 MILLIVOLTS
FUNDAMENTAL (MAX): 40.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.1 OUNCE-INCH

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SYNCHRO, CONTROL TRANSMITTER, TYPE 23CX4D

MIL-S-20708/45



CIRCUIT DIAGRAM

CHARACTERISTICS:

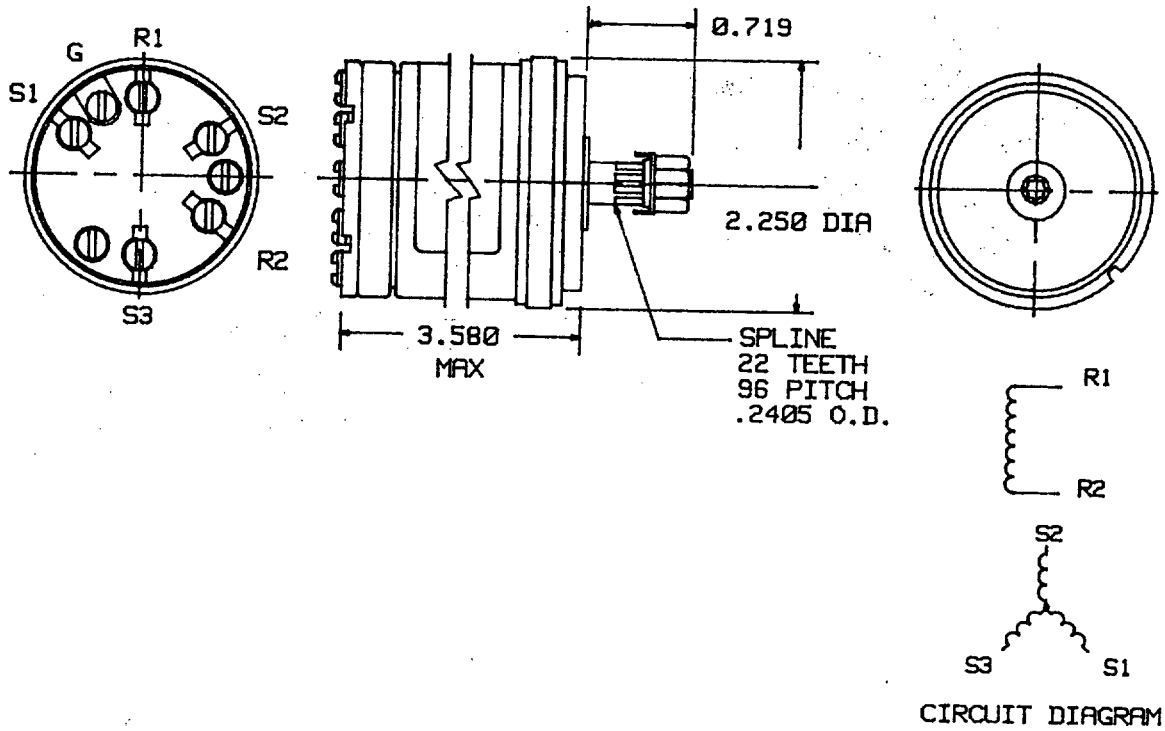
PART NUMBER: M20708/45-01D

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 245.0 MILLIAMPS
PRIMARY POWER (NOMINAL): 1.7 WATT
IMPEDANCE:
ZRO: 525 OHMS
ZSS: 23.8 OHMS
IMPEDANCE ANGLE:
ZRO: 85.8 DEGREES
ZSS: 44.5 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 1.0 (LEAD) DEGREE
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 48.0 MILLIVOLTS
FUNDAMENTAL (MAX): 32.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, CONTROL TRANSFORMER, TYPE 23CT4C

MIL-S-20708/46



CIRCUIT DIAGRAM

CHARACTERISTICS:

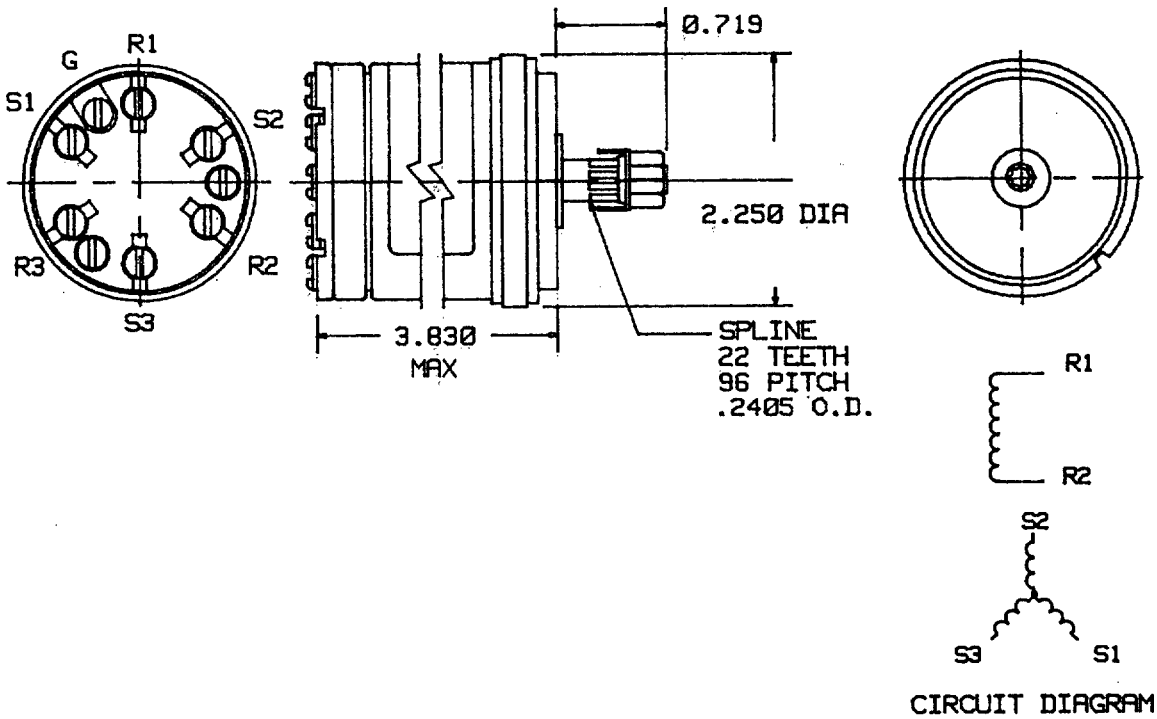
PART NUMBER: M20708/46-01C

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 5.7 MILLIAMPS
PRIMARY POWER (NOMINAL): 0.05 WATT
IMPEDANCE:
ZSO: 15,050 OHMS
ZRS: 1,150 OHMS
IMPEDANCE ANGLE:
ZSO: 82.8 DEGREES
ZRS: 51.0 DEGREES
TRANSFORMATION RATIO: 0.735
PHASE SHIFT: 2.3 (LEAD) DEGREE
ELECTRICAL ERROR (MAX): 6 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 45.0 MILLIVOLTS
FUNDAMENTAL (MAX): 20.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, CONTROL DIFFERENTIAL TRANSMITTER, TYPE 23CDX4C

MIL-S-20708/47



CIRCUIT DIAGRAM

CHARACTERISTICS:

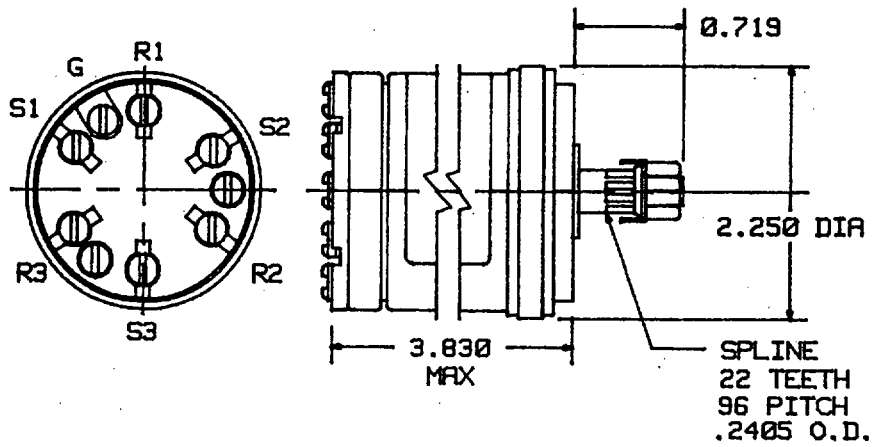
PART NUMBER: M20708/47-01C

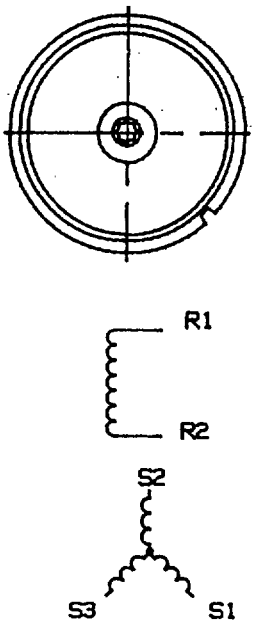
FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 285 MILLIAMPS
PRIMARY POWER (NOMINAL): 2.20 WATT
IMPEDANCE:
ZSO: 307.5 OHMS
ZRS: 43.5 OHMS
IMPEDANCE ANGLE:
ZSO: 84.0 DEGREES
ZRS: 47.5 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 2.0 (LEAD) DEGREE
ELECTRICAL ERROR (MAX): 7 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 60.0 MILLIVOLTS
FUNDAMENTAL (MAX): 30.0 MILLIVOLTS
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, TORQUE DIFFERENTIAL TRANSMITTER, TYPE 23TDX4C

MIL-S-20708/48





CIRCUIT DIAGRAM

CHARACTERISTICS:

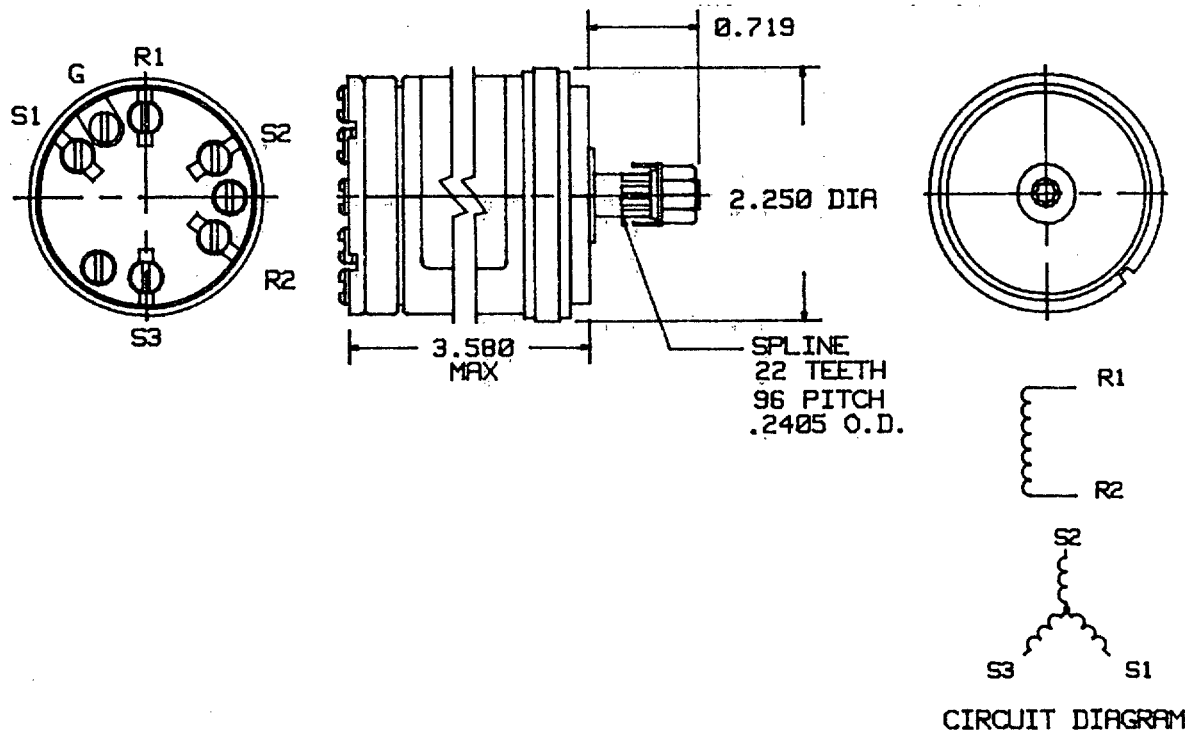
PART NUMBER: M20708/48-01C

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 90 VOLTS
PRIMARY CURRENT: 866 MILLIAMPS
PRIMARY POWER (NOMINAL): 7.0 WATT
IMPEDANCE:
ZSO: 101.0 OHMS
ZRS: 13.0 OHMS
IMPEDANCE ANGLE:
ZSO: 84.0 DEGREES
ZRS: 50.0 DEGREES
TRANSFORMATION RATIO: 1.154
PHASE SHIFT: 2.0 (LEAD) DEGREE
ELECTRICAL ERROR (MAX): 8 MINUTES
TORQUE GRADIENT (MIN): 0.16 OUNCE-INCH/DEGREE
FRICTION TORQUE (MAX): 0.2 OUNCE-INCH

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 23TRX4A

MIL-S-20708/50



CIRCUIT DIAGRAM

CHARACTERISTICS:

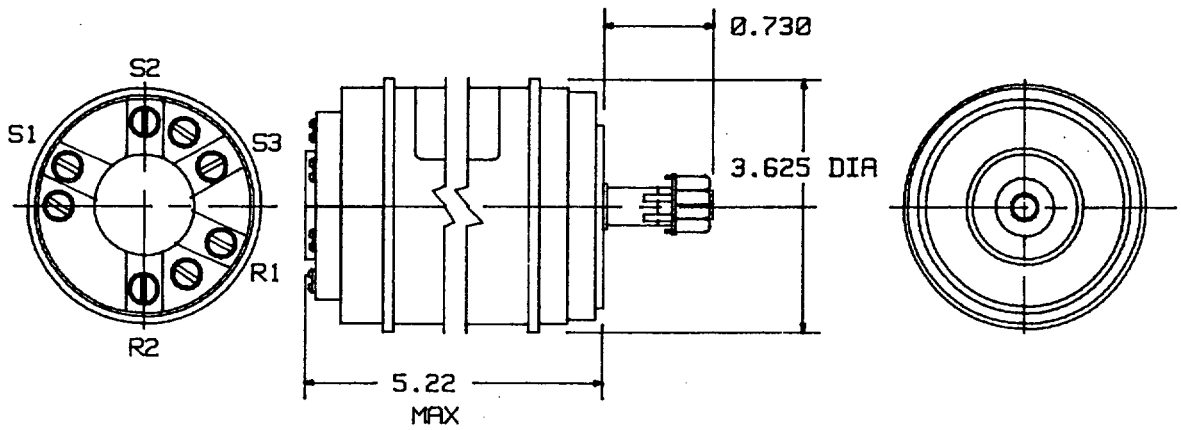
PART NUMBER: M20708/50-01A

FREQUENCY: 400 Hz
PRIMARY VOLTAGE (NOMINAL): 115 VOLTS
PRIMARY CURRENT: 720 MILLIAMPS
PRIMARY POWER (NOMINAL): 4.6 WATTS
IMPEDANCE:
ZSS: 7.3 OHMS
IMPEDANCE ANGLE:
ZSS: 44.0 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 2.0 (LEAD) DEGREE
ELECTRICAL ERROR (MAX): 6 MINUTES
RECEIVER ERROR (MAX): 45 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 150.0 MILLIVOLTS
FUNDAMENTAL (MAX): 30.0 MILLIVOLTS
TORQUE GRADIENT (MIN): 0.25 OUNCE-INCH/DEGREE

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SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 37TRX4A

MIL-S-20708/70



CHARACTERISTICS:

PART NUMBER: M20708/70-001

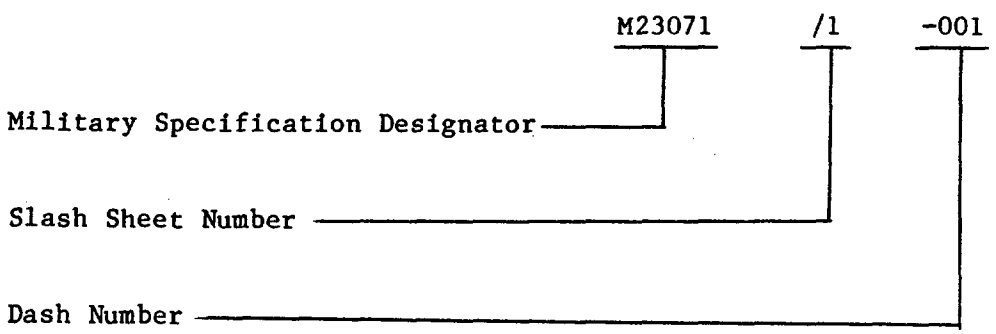
FREQUENCY: 400 Hz
PRIMARY VOLTAGE: 115 VOLTS
PRIMARY CURRENT (MAX): 1885 MILLIAMPS
PRIMARY POWER (MAX): 20.0 WATTS
IMPEDANCE:
ZSS: 3.0-4.0 OHMS / 66-70 DEGREES
TRANSFORMATION RATIO: 0.783
PHASE SHIFT: 1.0 (LEAD) DEGREES MAX
RECEIVER ERROR (MAX): 36.0 MINUTES
ELECTRICAL ERROR (MAX): 8.0 MINUTES
NULL VOLTAGE:
TOTAL (MAX): 170 MILLIVOLTS
FUNDAMENTAL (MAX): 35.0 MILLIVOLTS
TORQUE GRADIENT (MIN): 0.9 OUNCE-INCH/DEGREE

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BLOWERS

SCOPE: This section covers alternating current (ac) fractional horsepower miniature BLOWERS having specific environmental characteristics. These blowers are intended for uses requiring them for cooling communication, special military equipment, and other electronic equipment. These blowers shall be capable of withstanding extreme service and environmental conditions such as shock, high and low temperatures, vibration, humidity, and fungus.

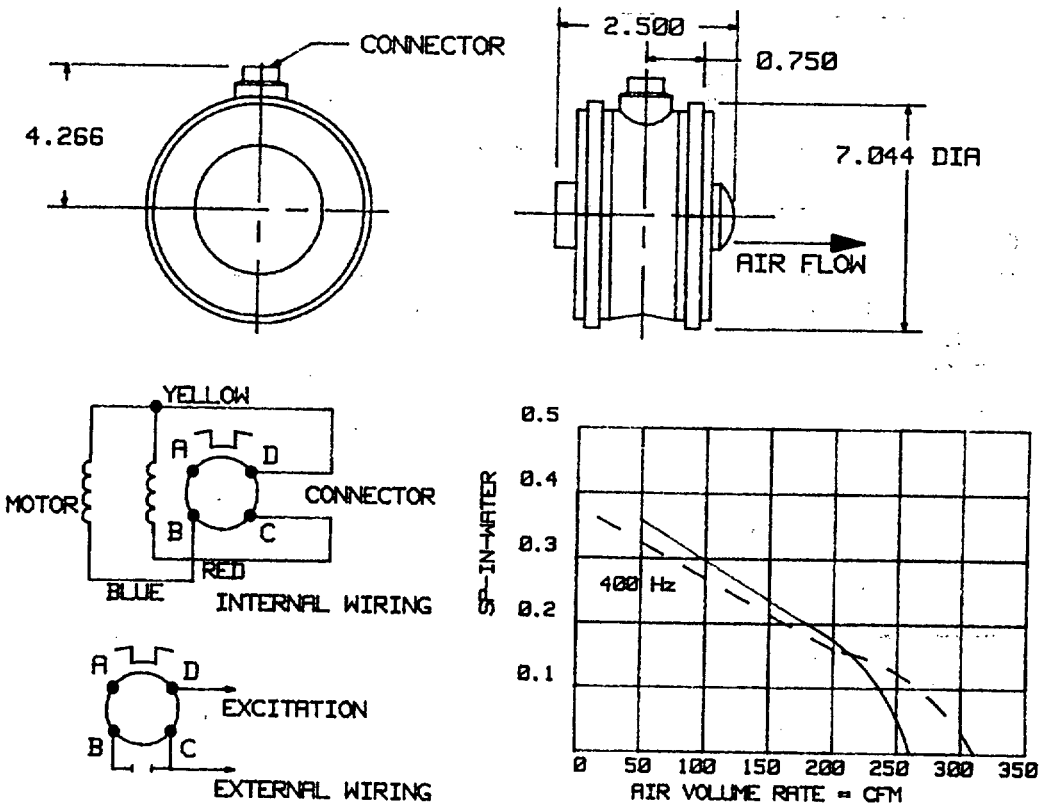
PART NUMBER EXAMPLE: M23071/1-001



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:29Z
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BLOWER, TUBEAXIAL

MIL-B-23071/1



PART NUMBER: MIL-B-23071/1-D-1

NOTES:

1. CONNECTOR: MS3112-8-4P
2. CAPACITOR: NOT SUPPLIED WITH BLOWER. THE CAPACITANCE VALUE AND WORKING VOLTAGE WILL DEPEND ON THE SPECIFIC BLOWER TYPES AS FOLLOWS: M23071/1-001, 2.0 MFD, 220 WVAC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:29Z
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BLOWER, TUBAXIAL

MIL-B-23071/1

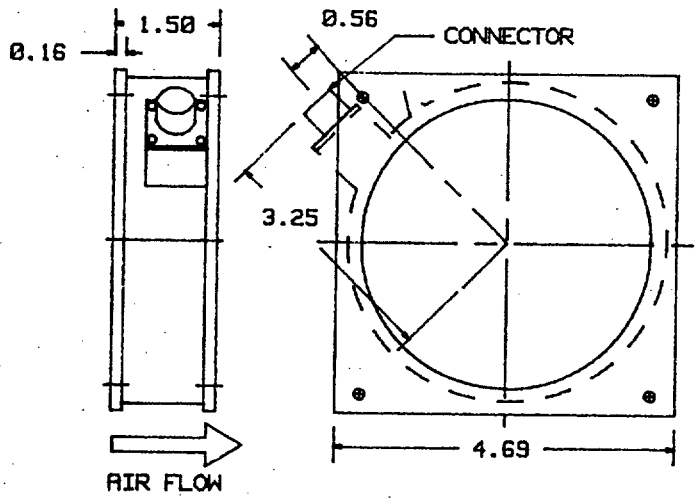
TABLE I. CHARACTERISTICS.

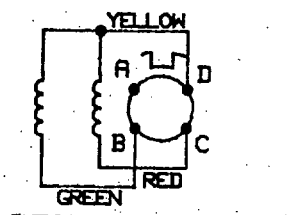
PART NUMBER:	M23071/1-001
BLOWER WEIGHT:	2 LBS MAX
EXCITATION (NOMINAL): VOLTAGE FREQUENCY PHASE DUTY CYCLE SPEED	 115 VOLTS, RMS ± 10 PERCENT 60 Hz ± 5 PERCENT 1 CONTINUOUS 3200 RPM
EXCITATION (RATED): VOLTAGE FREQUENCY	 115 VOLTS, RMS ± 2 PERCENT 60 Hz ± 1 PERCENT
POWER CONSUMPTION AT RATED EXCITATION: CURRENT, RUNNING CURRENT, LOCK ROTOR POWER, INPUT	 0.25 AMPS MAX 0.36 AMPS MAX 26 WATTS MAX
OUTPUT PERFORMANCE AT RATED EXCITATION: AIR VOLUME, ZERO SP AT .075 AIR VOLUME, RATED SP AT .075	 265 CFM MIN 190 CFM MIN AT 0.18 INCHES WATER
TEMPERATURE RISE, WINDING:	60°C MAX
GENERATED VIBRATION:	2.8 LBS-RMS MAX
ENVIRONMENTAL TEST LIMITS: COLD START TEST EXTENDED LIFE TEST	 -54°C ± 5 PERCENT 1000 HRS MIN AT 100°C
LIFE EXPECTANCY:	55000 HRS MTBF MIN AT 50°C 40000 HRS MTBF MIN AT 65°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:29Z
Check the source to verify that this is the current version before use.

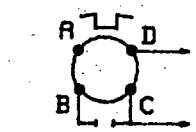
BLOWER, TUBEAXIAL

MIL-B-23071/7

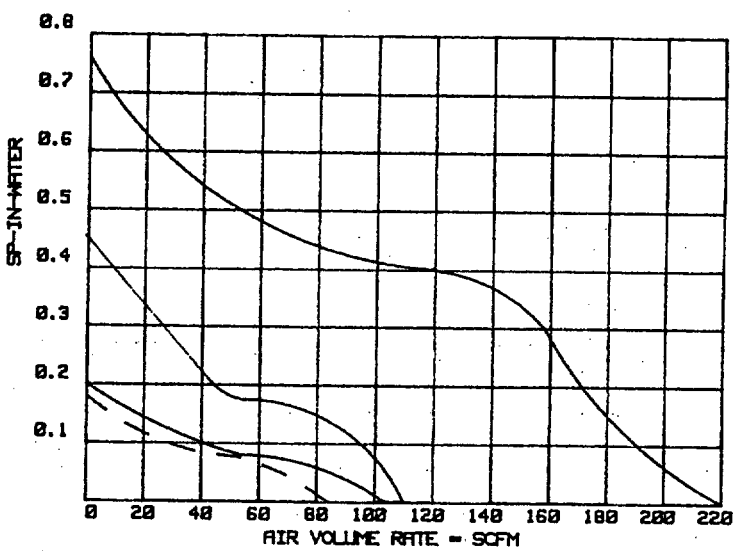




INTERNAL MOTOR WIRING



EXTERNAL CONNECTOR WIRING



NOTES:

1. CONNECTOR: MS3112E-8-4P
2. CAPACITOR: NOT SUPPLIED WITH BLOWER. THE CAPACITANCE VALUE AND WORKING VOLTAGE WILL DEPEND ON THE SPECIFIC BLOWER TYPES AS FOLLOWS: M23071/7-001, 1.0 MFD, 220 WVAC; M23071/7-002, 0.22 MFD, 237 WVAC; M23071/7-003, 1.0 MFD, 220 WVAC.

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BLOWER, TUBAXIAL

MIL-B-23071/7

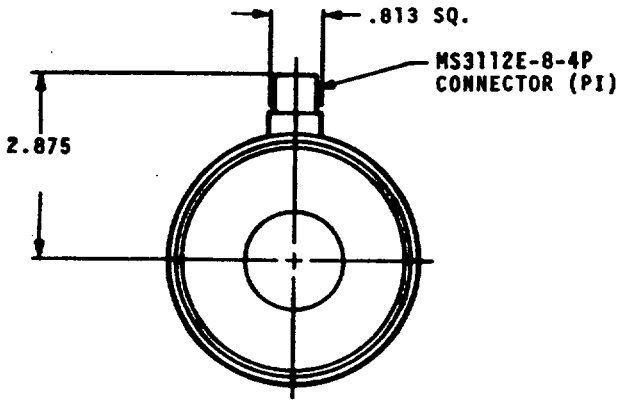
TABLE I. CHARACTERISTICS.

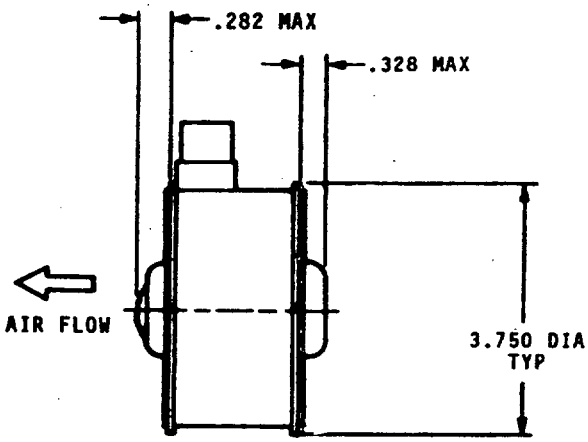
PART NUMBER:	M23071/7-			
DASH NUMBER:	001	002	003	
BLOWER WEIGHT:	1.5 LBS (MAX)			
EXCITATION (NOMINAL):				
VOLTAGE	115 V, RMS			
FREQUENCY	60	400	60/400	Hz
PHASE	1			
DUTY CYCLE	CONTINUOUS			
SPEED	3200	5500	2500/2800	RPM
EXCITATION (RATED):				
VOLTAGE	115 V, RMS			
FREQUENCY	60	400	60/400	Hz
POWER CONSUMPTION AT RATED EXCITATION:				
CURRENT, RUNNING	0.12	0.77	0.22/0.50	AMPS (MAX)
CURRENT, LOCK ROTOR	0.17	0.99	0.22/0.50	AMPS (MAX)
POWER, INPUT	14	40	17.0/39.0	WATTS (MAX)
OUTPUT PERFORMANCE AT RATED EXCITATION:				
AIR VOLUME, ZERO SP AT .075	110	180	85/85	CFM (MIN)
AIR VOLUME, RATED SP AT .075	76	160	60/60	CFM (MIN) AT 1.5 INCH WATER
TEMPERATURE RISE, WINDING:	44	55	32/60	°C (MAX)
GENERATED VIBRATION:	2.8 LBS-RMS (MAX)			
ENVIRONMENTAL TEST LIMITS:				
COLD START TEST	-54°C			
EXTENDED LIFE TEST	1000 HRS MIN AT 100°C			
LIFE EXPECTANCY:	65000	20000	20000	HRS (MIN @50°C)
	50000	15000	15000	HRS (MIN @65°C)

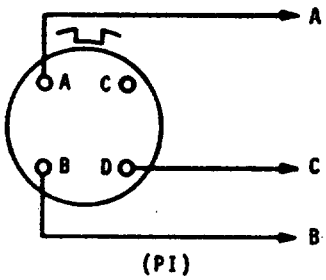
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:29Z
Check the source to verify that this is the current version before use.

BLOWER, TUBEAXIAL

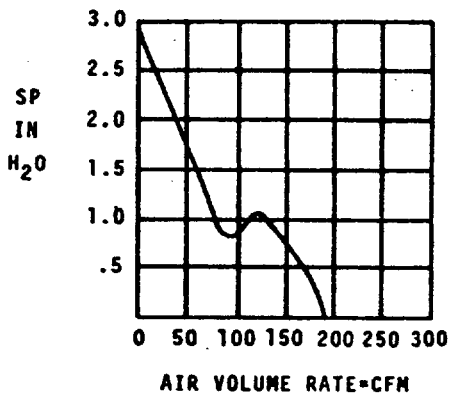
MIL-B-23071/8







EXTERNAL CONNECTIONS



NOTE:

CAPACITOR: 2.0 MFD, 220 WVAC RATED, NOT SUPPLIED.

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BLOWER, TUBAXIAL

MIL-B-23071/8

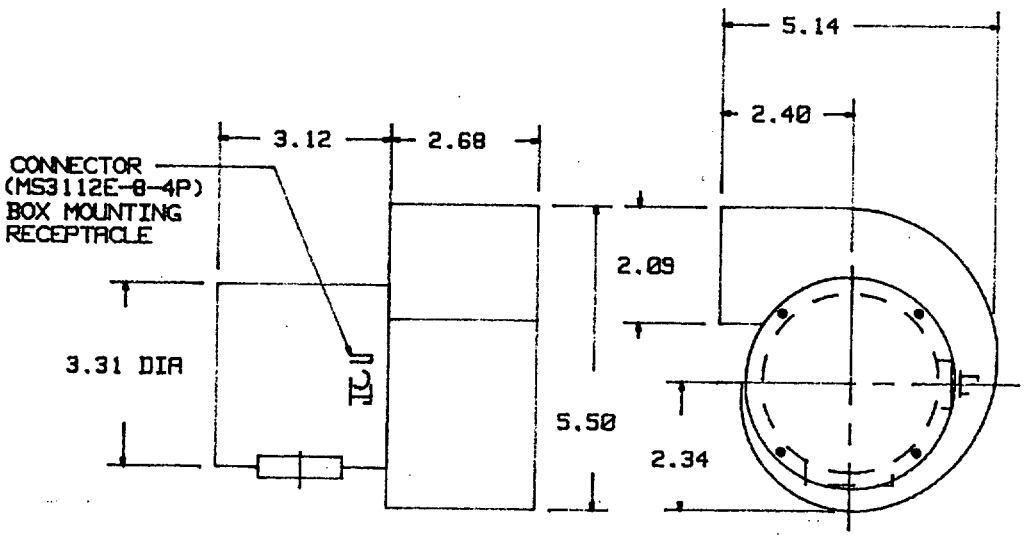
TABLE I. CHARACTERISTICS.

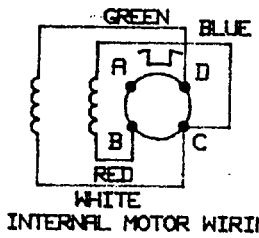
PART NUMBER:	M23071/8-001
BLOWER WEIGHT:	1.25 LBS MAX
EXCITATION (NOMINAL): VOLTAGE FREQUENCY PHASE DUTY CYCLE SPEED	 200 VOLTS, RMS ± 10 PERCENT 400 Hz ± 5 PERCENT 3 CONTINUOUS 11000 RPM
EXCITATION (RATED): VOLTAGE FREQUENCY	 200 VOLTS, RMS ± 2 PERCENT 400 Hz ± 1 PERCENT
POWER CONSUMPTION AT RATED EXCITATION: CURRENT, RUNNING CURRENT, LOCK ROTOR POWER, INPUT	 0.32 AMPS MAX 0.96 AMPS MAX 69 WATTS MAX
OUTPUT PERFORMANCE AT RATED EXCITATION: AIR VOLUME, ZERO SP AT .075 AIR VOLUME, RATED SP AT .075	 174 CFM MIN 141 CFM MIN AT 0.7 INCHES WATER
TEMPERATURE RISE, WINDING:	55°C MAX
GENERATED VIBRATION:	2.8 LBS-RMS MAX
ENVIRONMENTAL TEST LIMITS: COLD START TEST EXTENDED LIFE TEST	 -54°C ± 5 PERCENT 1000 HRS MIN AT 125°C
LIFE EXPECTANCY:	50000 HRS MTBF MIN AT 50°C 40000 HRS MTBF MIN AT 65°C

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:29Z
Check the source to verify that this is the current version before use.

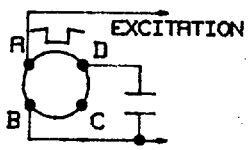
BLOWER, CENTRIFUGAL

MIL-B-23071/3

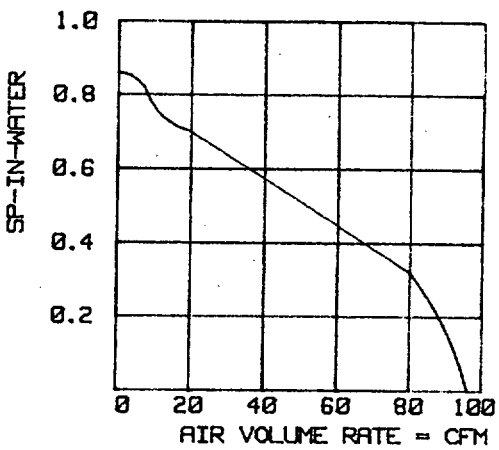




INTERNAL MOTOR WIRING



EXTERNAL CONNECTOR WIRING



NOTES:

1. CONNECTOR -MS3112E-8-4P
2. CAPACITOR: -4.0 MFD., 220 WVAC RATED.
3. OUTLET ORIENTATION WITH RESPECT TO THE MOUNTING FLATS: THE BLOWER SHALL BE SUPPLIED WITH THE DISCHARGE OUTLET DIRECTION AS SHOWN. THE BLOWER SHALL BE CAPABLE OF DISCHARGE DIRECTIONS 90°, 180°, AND 270° FROM THAT SHOWN.

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BLOWER, CENTRIFUGAL

MIL-B-23071/3

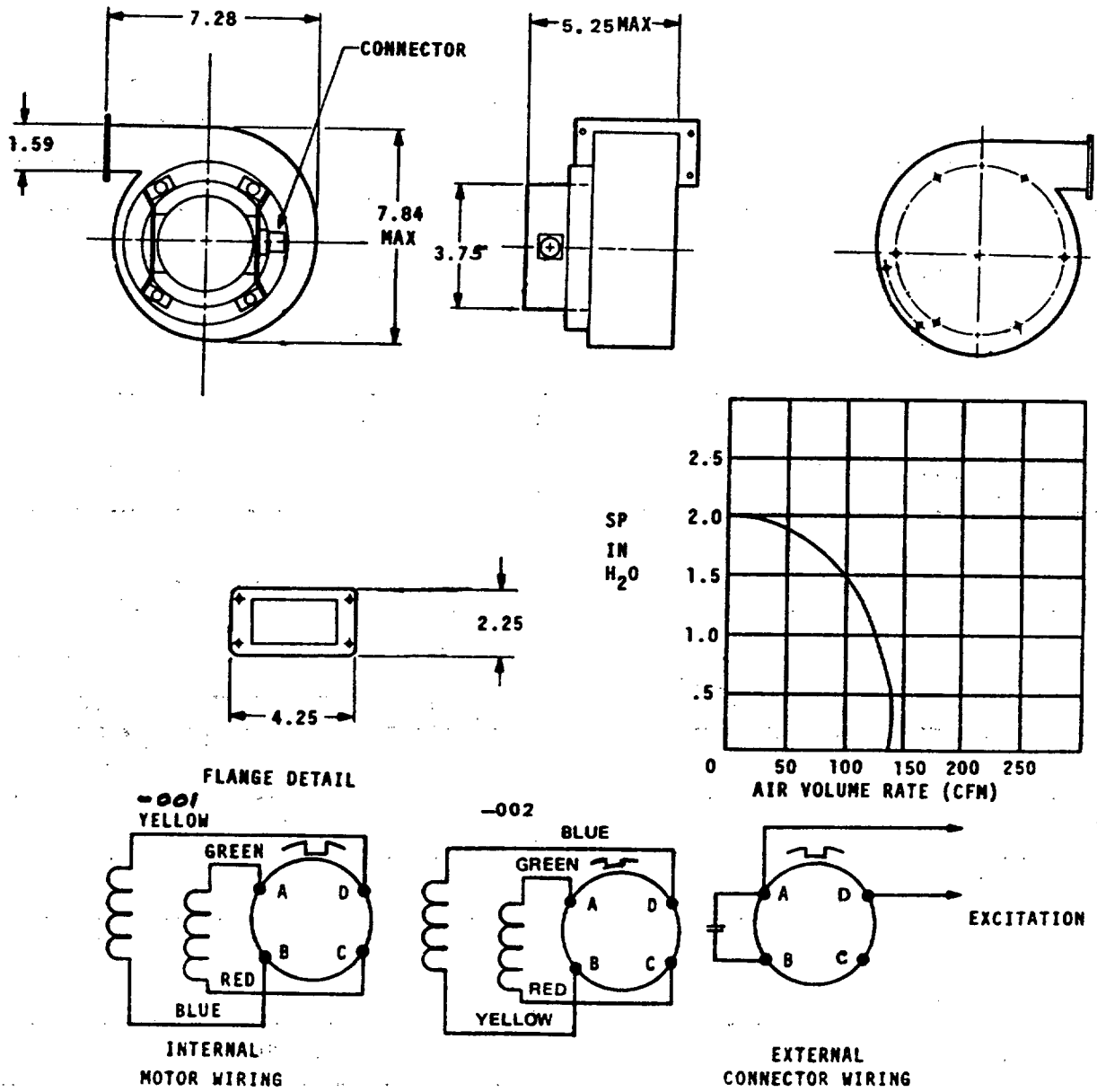
TABLE I. CHARACTERISTICS.

PART NUMBER:	M23071/3-001
BLOWER WEIGHT:	4 LBS MAX
EXCITATION (NOMINAL): VOLTAGE FREQUENCY PHASE DUTY CYCLE SPEED	 115 VOLTS, RMS ± 10 PERCENT 60 Hz ± 5 PERCENT 1 CONTINUOUS 3300 RPM
EXCITATION (RATED): VOLTAGE FREQUENCY	 115 VOLTS, RMS ± 2 PERCENT 60 Hz ± 1 PERCENT
POWER CONSUMPTION AT RATED EXCITATION: CURRENT, RUNNING CURRENT, LOCK ROTOR POWER, INPUT	 0.25 AMPS MAX 0.50 AMPS MAX 50 WATTS MAX
OUTPUT PERFORMANCE AT RATED EXCITATION: AIR VOLUME, ZERO SP AT .075 AIR VOLUME, RATED SP AT .075	 93 CFM MIN 70 CFM MIN AT 0.40 INCHES WATER
TEMPERATURE RISE, WINDING:	40°C MAX
GENERATED VIBRATION:	2.8 LBS-RMS MAX
LINE VOLTAGE AND FREQUENCY EXCITATION LIMITS: THE MAXIMUM DECREASE IN SHAFT SPEED AT ANY COMBINATION OF NOMINAL EXCIATION (IN ACCORDANCE WITH TABLE VI OF MIL-B-23071) SHALL NOT BE GREATER THAN 20 PERCENT OF THE REFERENCE SPEED AT RATED EXCITATION.	
ENVIRONMENTAL TEST LIMITS: COLD START TEST EXTENDED LIFE TEST	 -54°C ± 5 PERCENT 1000 HRS MIN AT 100°C
LIFE EXPECTANCY:	55000 HRS MTBF MIN AT 50°C 45000 HRS MTBF MIN AT 65°C

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BLOWER, CENTRIFUGAL

MIL-B-23071/9



NOTES:

1. CONNECTOR: MS3112E-8-4P. LOCATION OF CONNECTOR IS OPTIONAL WITHIN 80°.
2. CAPACITOR: 4.0 MFD., 220 WVAC RATED, NOT SUPPLIED WITH BLOWER.

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BLOWER, CENTRIFUGAL

MIL-B-23071/9

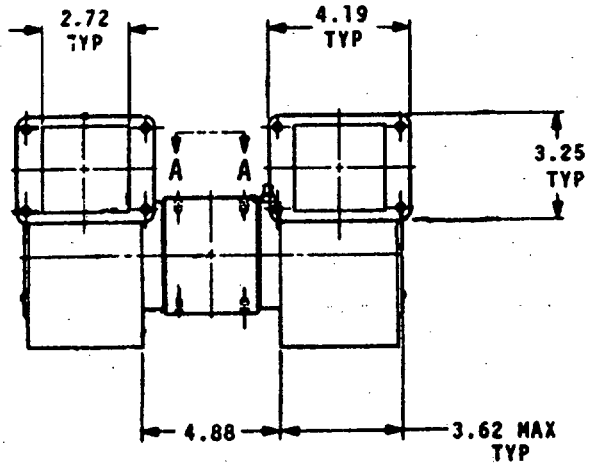
TABLE I. CHARACTERISTICS.

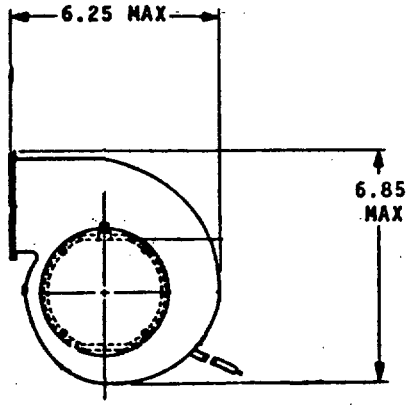
PART NUMBER:	M23071/9-		
DASH NUMBER:	001	002	
BLOWER WEIGHT:	1.5 LBS (MAX)		
EXCITATION (NOMINAL):			
VOLTAGE	115 V, RMS		
FREQUENCY	60 Hz		
PHASE	1		
DUTY CYCLE	CONTINUOUS		
SPEED	3200 RPM		
EXCITATION (RATED):			
VOLTAGE	115 V, RPM		
FREQUENCY	60 Hz		
POWER CONSUMPTION AT RATED EXCITATION:			
CURRENT, RUNNING	1.2 AMPS (MAX)		
CURRENT, LOCK ROTOR	2.4	2.9	AMPS (MAX)
POWER, INPUT	138 WATTS (MAX)		
OUTPUT PERFORMANCE AT RATED EXCITATION:			
AIR VOLUME, ZERO SP AT .075	135	130	CFM (MIN)
AIR VOLUME, RATED SP AT .075	101	98	CFM (MIN) AT 1.25 INCH WATER
TEMPERATURE RISE, WINDING:	35 °C (MAX)		
GENERATED VIBRATION:	2.8 LBS-RM		
ENVIRONMENTAL TEST LIMITS:			
COLD START TEST	-54°C		
EXTENDED LIFE TEST	1000 HRS MIN		
LIFE EXPECTANCY:	20000 HRS (MIN @50°C)		
	15000 HRS (MIN @65°C)		

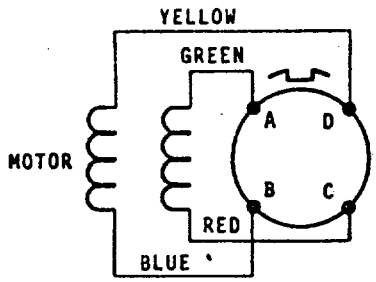
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Check the source to verify that this is the current version before use.

BLOWER, CENTRIFUGAL

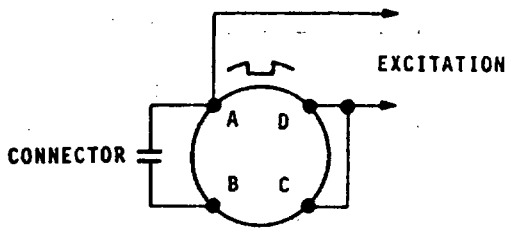
MIL-B-23071/10



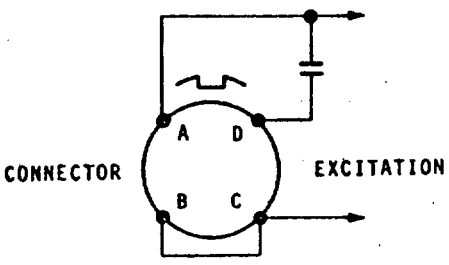




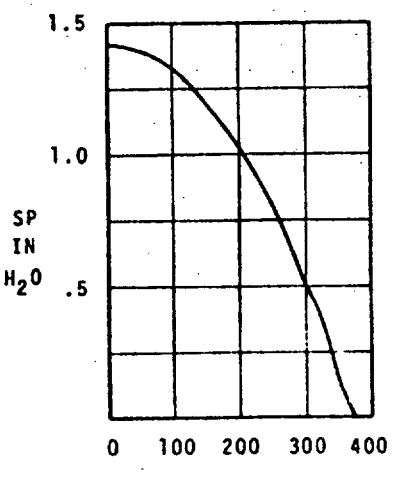
INTERNAL WIRING



EXTERNAL WIRING-CW



EXTERNAL WIRING-CCW



AIR VOLUME RATE-CFM

NOTES:
1. CONNECTOR: MS3112E-8-4P.
2. CAPACITOR: 2.5 MFD., 330 WVAC.

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BLOWER, CENTRIFUGAL

MIL-B-23071/10

TABLE I. CHARACTERISTICS.

PART NUMBER:	M23071/10-001
BLOWER WEIGHT:	20 LBS MAX
EXCITATION (NOMINAL): VOLTAGE FREQUENCY PHASE DUTY CYCLE SPEED	 115 VOLTS, RMS ± 10 PERCENT 60 Hz ± 5 PERCENT 1 CONTINUOUS 3300 RPM
EXCITATION (RATED): VOLTAGE FREQUENCY	 115 VOLTS, RMS ± 2 PERCENT 60 Hz ± 1 PERCENT
POWER CONSUMPTION AT RATED EXCITATION: CURRENT, RUNNING CURRENT, LOCK ROTOR POWER, INPUT	 1.65 AMPS MAX 3.60 AMPS MAX 150 WATTS MAX
OUTPUT PERFORMANCE AT RATED EXCITATION: AIR VOLUME, ZERO SP AT .075 AIR VOLUME, RATED SP AT .075	 360 CFM MIN 270 CFM MIN AT 0.68 INCHES WATER
TEMPERATURE RISE, WINDING:	35°C MAX
GENERATED VIBRATION:	2.8 LBS-RMS MAX
LINE VOLTAGE AND FREQUENCY EXCITATION LIMITS: THE MAXIMUM DECREASE IN SHAFT SPEED AT ANY COMBINATION OF NOMINAL EXCIATION (IN ACCORDANCE WITH TABLE VI OF MIL-B-23071) SHALL NOT BE GREATER THAN 25 PERCENT OF THE REFERENCE SPEED AT RATED EXCITATION.	
ENVIRONMENTAL TEST LIMITS: COLD START TEST EXTENDED LIFE TEST	 -54°C ± 5 PERCENT 1000 HRS MIN AT 100°C
LIFE EXPECTANCY:	15000 HRS MTBF MIN AT 50°C 10000 HRS MTBF MIN AT 65°C

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ACOUSTICAL PARTS

SCOPE: This section includes several ACOUSTICAL PARTS including handsets, headsets, microphones, headsets-microphones, and loudspeakers, respectively. These items are covered by different controlling documnets. For the particular part number, type, controlling document and characteristic see the page specified in Table I below.

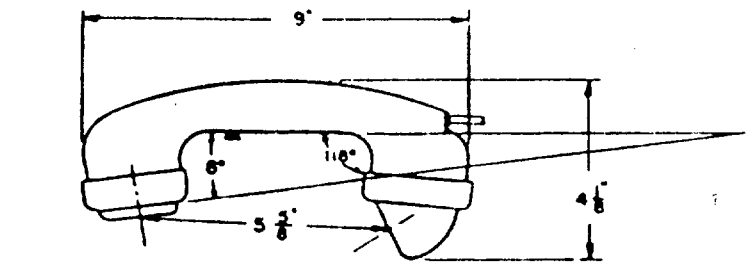
TABLE I. PART, TYPE, AND PAGE NUMBERS.

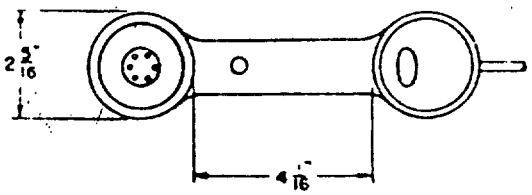
PART	TYPE	PAGE NUMBER
HANDSET	H-203/U	310.1
	H-200/U	310.2
HEADSET	NT-49985A	320.1
	NT-49507B	320.2
	H-216/U	320.3
	H-43B/U	320.4
MICROPHONE	M-105/U	330.1
	M-109A/U	330.2
	M-156/U	330.3
	M-101/AIC	330.4
	M-87/AIC	330.5
HEADSET-MICROPHONE	H-62/U	340.1
	H-170/U	340.2
	H-172/U	340.3
LOUDSPEAKER	LS-455/U	350.1
	LS-445/U	350.2
	LS-166/U	350.3
	LS-454/U	350.3
	NT-491814	350.4
	LS-169/U	350.5
	LS-180/U	350.6
	LS-474/U	350.7
	NT-49546	350.8

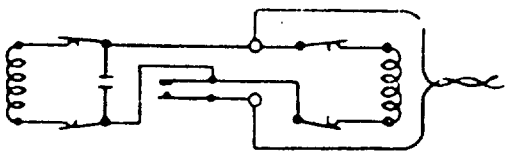
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HANDSET

H-203/U







HANDSET WIRING TYPE H-203/U

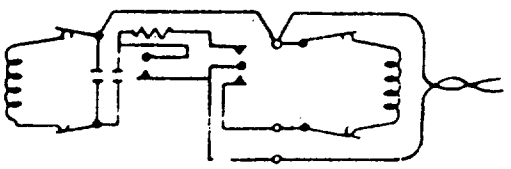


TABLE I. MINIMUM RESPONSE

FREQUENCY BAND C.P.S.	NO. SETS IN PARALLEL ON LINE		
	TWO H-200/U	TWO H-203/U	TEN H-200/U
750 - 1250	96	98	90
1250 - 1750	103	105	97
1800 - 2600	103	102	94
500 - 2500	101	102	94

PART NUMBER: H-203/U

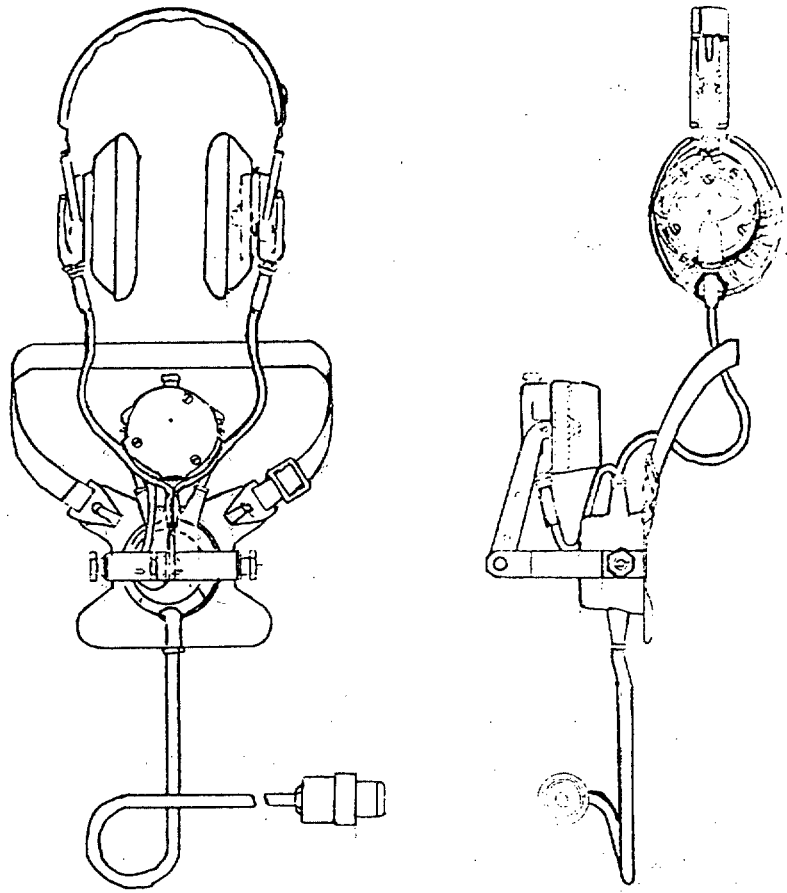
DESCRIPTION: SOUND POWERED TELEPHONE HANDSET, FOR USE IN HIGH INTELLIGIBILITY, RUGGEDNESS, RELIABILITY, SIMPLICITY, AND EASE OF MAINTENANCE AND OPERATION. THE HANDSET IS PROVIDED WITH A NONKINKING RETRACTABLE CORD, 1 FOOT LENGTH RETRACTED, 5 FEET EXTENDED.

CONTROLLING DOCUMENT: MIL-T-15514

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HANDSET

H-200/U



PART NUMBER: H-200/U

DESCRIPTION: SOUND POWERED TELEPHONE HEADSET-CHEST SET, FOR USE IN HIGH INTELLIGIBILITY, RUGGEDNESS, RELIABILITY, SIMPLICITY, AND EASE OF MAINTENANCE AND OPERATION.

FREQUENCY RANGE: 500 - 2500 C.P.S.

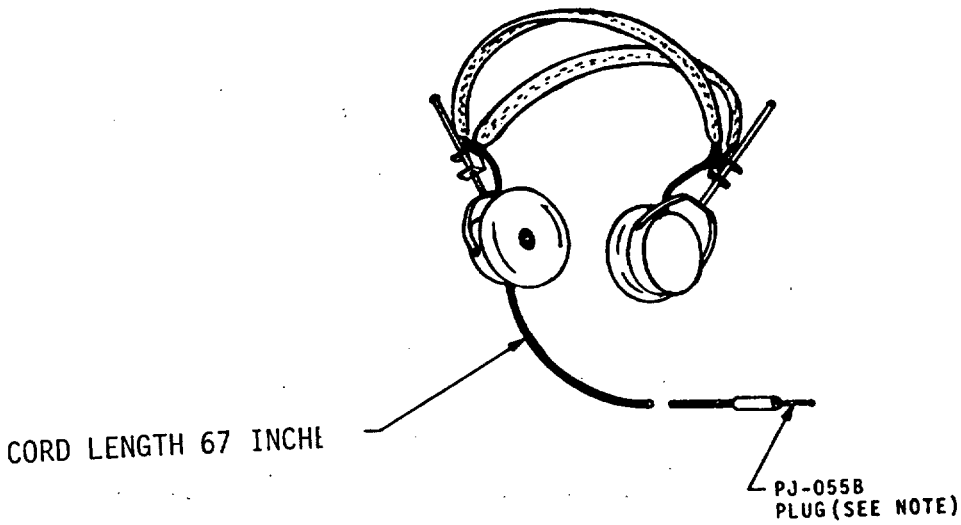
SEE TABLE I. PAGE 310.1

CONTROLLING DOCUMENT: MIL-T-15514 AND DRAWING 815-1853010.

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HEADSET

NT-49985A



PART NUMBER: NT-49985A

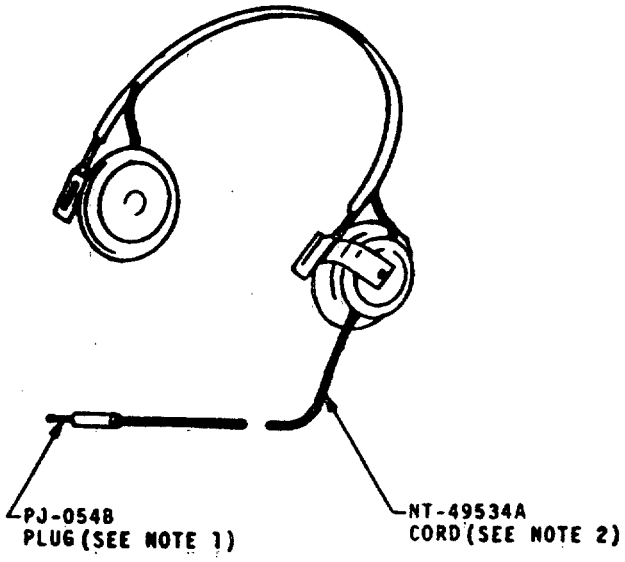
DESCRIPTION: HEADSET, MAGNETIC, GENERAL PURPOSE.
EARPHONE:
IMPEDENCE: 600 OHMS
FREQUENCY RANGE: 300 - 4000 Hz

NOTE:
MATES WITH JACKS PER MIL-J-641 MATING TABLE.

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Check the source to verify that this is the current version before use.

HEADSET

NT-49507B



PART NUMBER: NT-49507B

DESCRIPTION: HEADSET, ADJUSTABLE BAND, FOR USE UNDER COMBAT HELMETS AND FOR RUGGED FIELD CONDITIONS; SALT SPRAY RESISTANT.

EARPHONE:

IMPEDENCE: 600 OHMS (2 X 30 IN SERIES)

FREQUENCY RANGE: 200 - 5000 Hz

CONTROLLING DOCUMENT: RE-D268803

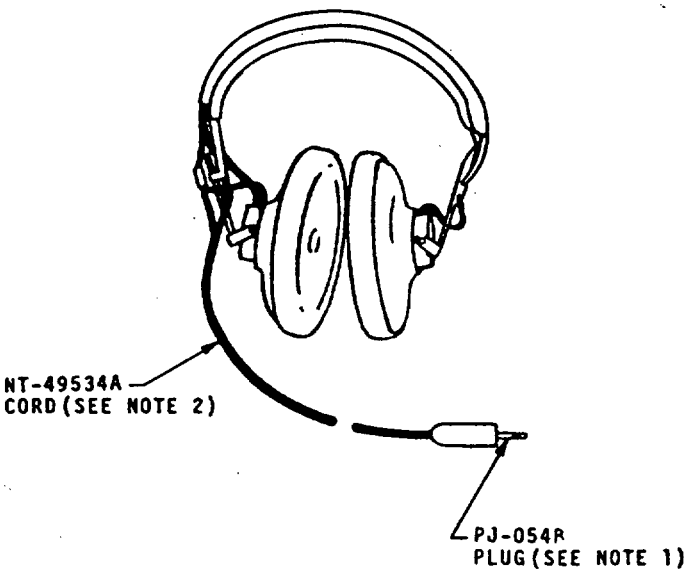
NOTES:

1. MATES WITH JACKS PER MIL-J-641 MATING TABLE.
2. CORD LENGTH: 9 INCHES; EXTENSION CORD AVAILABLE, 65 INCHES LONG, NT-49534 PER MIL-C-3849, TERMINATING IN PJ-054B.

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HEADSET

H-216/U



PART NUMBER: H-216/U

DESCRIPTION: HEADSET, ADJUSTABLE HEADBAND, FOR SONAR USE OR WHERE EXTENDED FREQUENCY RESPONSE IS REQUIRED.

EARPHONE:
IMPEDENCE: 600 OHMS (2 X 300 IN SERIES)
FREQUENCY RANGE: 100 - 7000 Hz

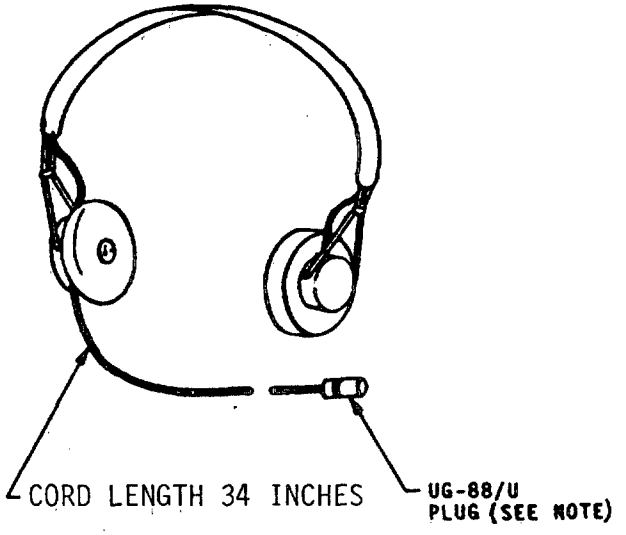
CONTROLLING DOCUMENT: MIL-H-55353

NOTES:
1. MATES WITH JACKS PER MIL-J-641 MATING TABLE.
2. CORD LENGTH: 7 INCHES; EXTENSION CORD AVAILABLE, 65 INCHES LONG, NT-49534 PER MIL-C-3849, TERMINATING IN PJ-054R.

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HEADSET

H-43B/U



PART NUMBER: H-43B/U

DESCRIPTION: HEADSET, MAGNETIC ADJUSTABLE HEADBAND, FOR USE WITH RADIAC SETS; MAY BE WORN UNDER COMBAT HELMETS.

EARPHONE:
- IMPEDENCE: 22,000 OHMS EACH, SERIES CONNECTED
- FREQUENCY RANGE: 300 - 3000 Hz
- FREQUENCY OF MAX RESPONSE: 1000 Hz NOMINAL

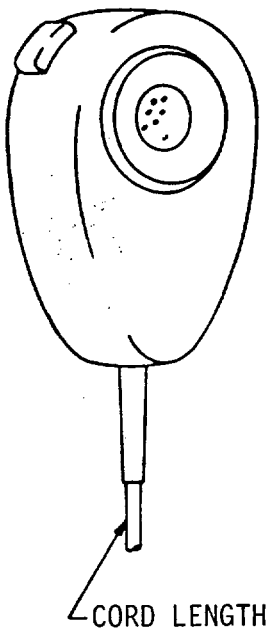
CONTROLLING DOCUMENT: REF 2688037

NOTE:
MATES WITH SERIES BNC RECEPTACLE, SOCKET CONTACT, PER MIL-C-39012/17

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MICROPHONE

M-105/U



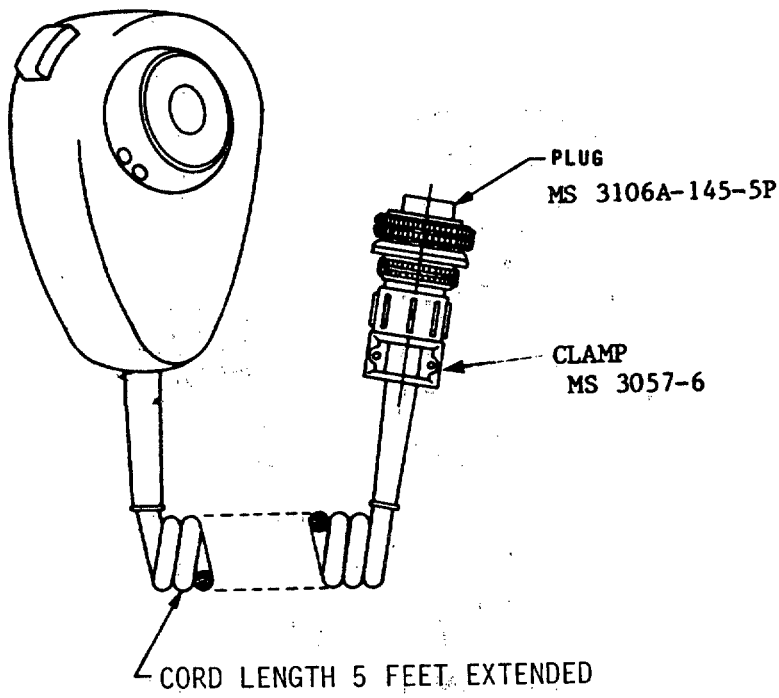
PART NUMBER: M-105/U

DESCRIPTION: MICROPHONE, DYNAMIC, GENERAL PURPOSE
MICROPHONE:
IMPEDENCE: 25,000 OHMS
FREQUENCY RANGE: 100 - 6000 Hz

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MICROPHONE

M-109A/U



PART NUMBER: M-109A/U

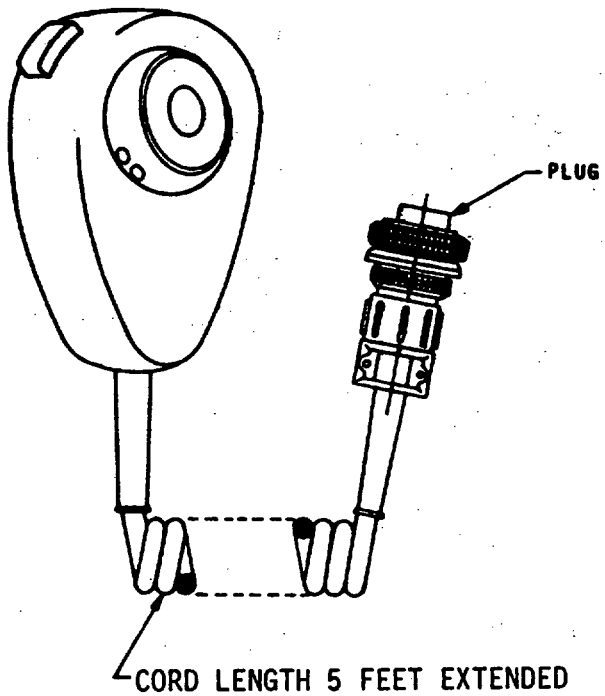
DESCRIPTION: MICROPHONE, NOISE CANCELLING; CONTAINS TRANSISTORIZED AMPLIFIER; LOW DISTORTION.

MICROPHONE:
RECOMMENDED AMPLIFIER LOAD: 30 OHMS
FREQUENCY RANGE: 1000 TO 5000 Hz
IM DISTORTION: 2% MAX
CONTROLLING DOCUMENT: RE-D2688038

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MICROPHONE

M-156/U



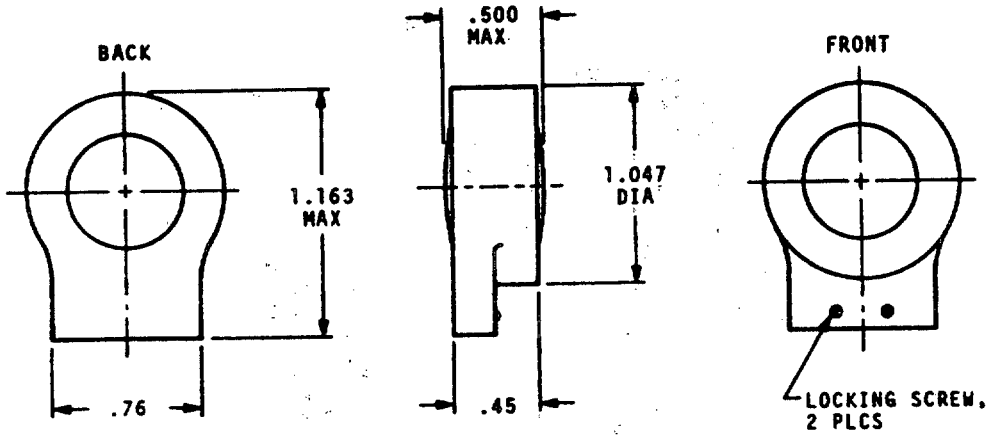
PART NUMBER: M-156/U

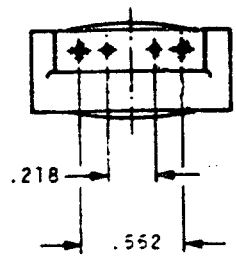
DESCRIPTION: MICROPHONE, DYNAMIC, NOISE CANCELLING
MICROPHONE:
IMPEDENCE: 50 OHMS
FREQUENCY RANGE: 100 TO 5000 Hz
CONTROLLING DOCUMENT: RE-D2688038

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MICROPHONE ELEMENT

M-101/AIC





PART NUMBER: M-101/AIC

DESCRIPTION: MICROPHONE ELEMENT, DYNAMIC, NOISE CANCELLING, FOR USE IN GAS MASKS, OXYGEN HELMENTS, ETC.; LOW DISTORTION MICROPHONE:

IMPEDENCE: 7 OHMS

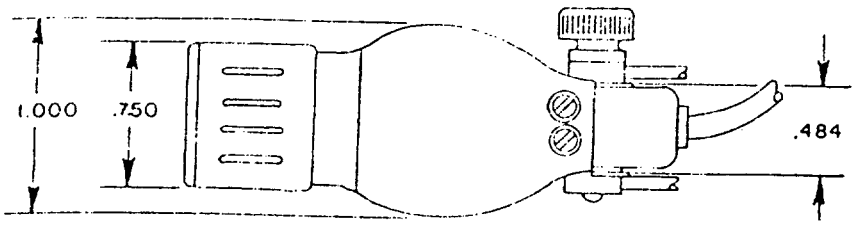
FREQUENCY RANGE: 200 TO 6000 Hz

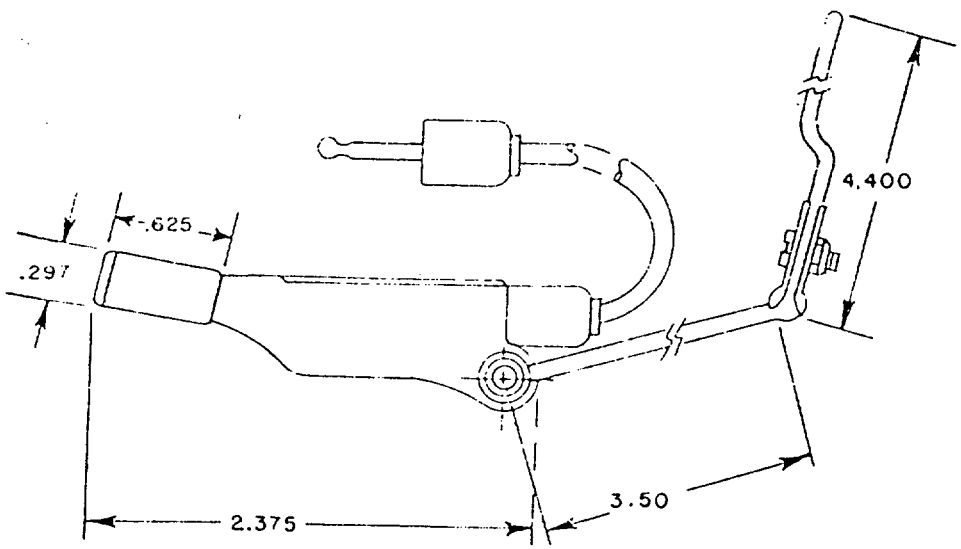
CONTROLLING DOCUMENT: MIL-M-26542/4

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MICROPHONES

M-87/AIC





PART NUMBER: M-87/AIC

DESCRIPTION: MICROPHONE, DYNAMIC, NOISE-CANCELLING, FOR USE ON A HEADBAND TYPE HEADSET AT LOW ALTITUDES, AN OXYGEN HELMET, OR TO PROVIDE COMMUNICATION UNDER NOISE CONDITIONS IN MILITARY AIRCRAFT.

IMPEDANCE: 4-6 OHMS

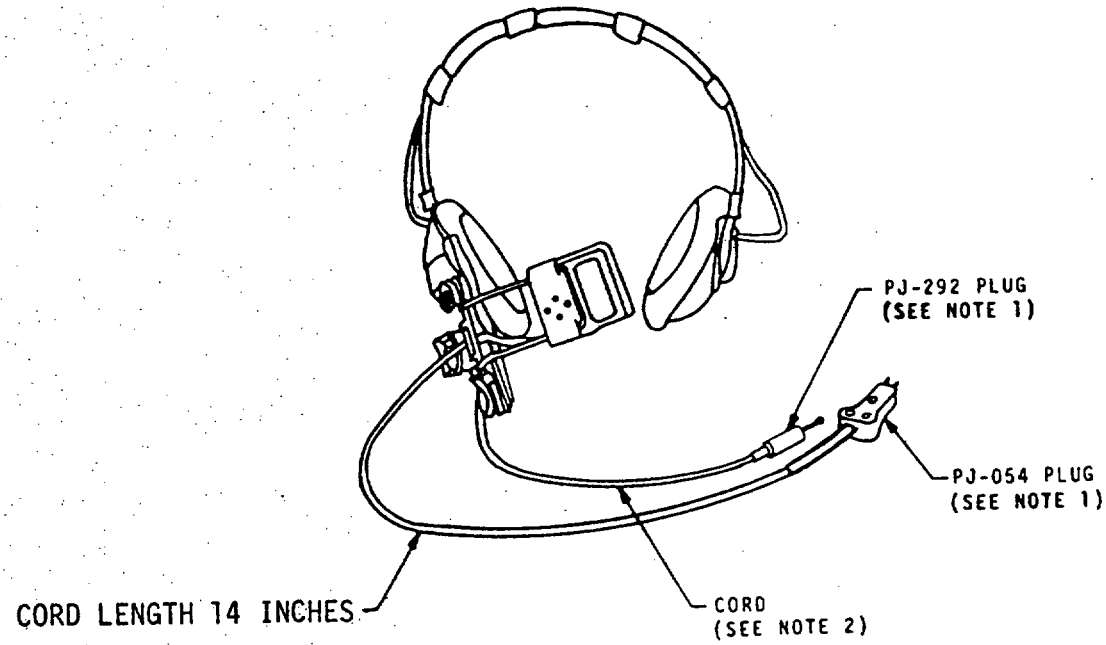
FREQUENCY RANGE: 200 TO 6000Hz

CONTROLLING DOCUMENT: MIL-M-26542/2

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HEADSET-MICROPHONE

H-62/U



PART NUMBER: H-62/U

DESCRIPTION: HEADSET-MICROPHONE ASSEMBLY, CARBON, NOISE CANCELLING MICROPHONE, ADJUSTABLE HEADBAND, FOR USE UNDER COMBAT HELMETS AND FOR RUGGED FIELD CONDITIONS; SALT AND GUN BLAST RESISTANT.

EARPHONES:
- IMPEDENCE: 600 OHMS (2 X 300 IN SERIES)
- FREQUENCY RANGE: 300 TO 4000 Hz

MICROPHONE:
- IMPEDENCE: 100 OHMS
- FREQUENCY RANGE: 300 TO 4000 Hz
- INPUT CURRENT: 50 mA

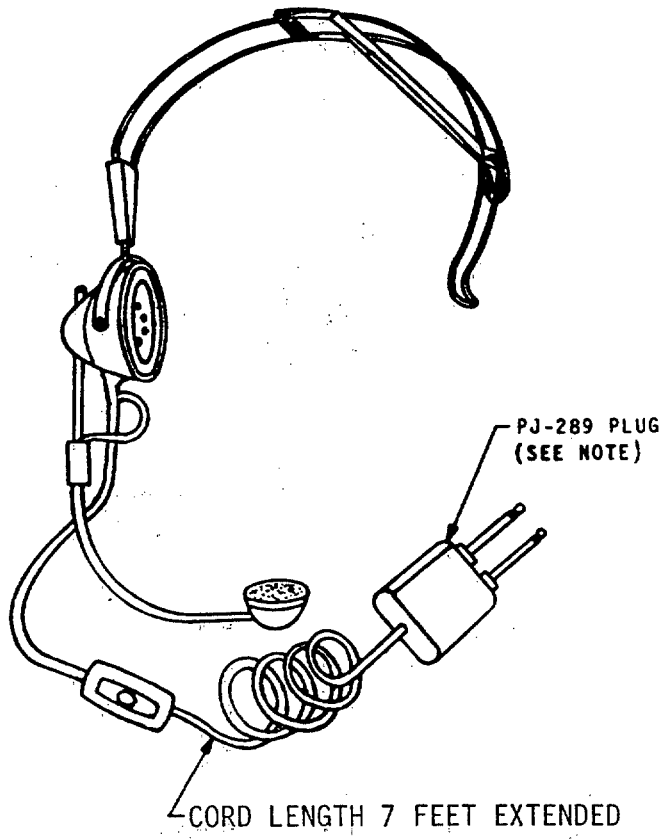
CONTROLLING DOCUMENT: NONE, THE ASSEMBLY MUST BE ORDERED AS SEPARATE PARTS.
- HEADSET: NT-49507B PER RE51D123
- BOOM: MX-1435/U PER RED 2680701
- MICROPHONE: M-51/UR PER SC-C-77023

NOTES:
1. MATES WITH JACKS PER MIL-J-641 MATING TABLE.
2. CORD LENGTH: 16 INCHES; EXTENSION CORD AVAILABLE 65 IN. LONG, NT-49534 PER MIL-C-3849, TERMINATING IN PJ-054B.

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HEADSET-MICROPHONE

H-170/U



PART NUMBER: H-170/U

DESCRIPTION: HEADSET-MICROPHONE, CARBON MICROPHONE, DYNAMIC EARPHONE, IN-LINE ON-OFF SWITCH.

EARPHONE:
- IMPEDENCE: 600 OHMS
- FREQUENCY RANGE: DC TO 2800 Hz

MICROPHONE:
- IMPEDANCE: 350 OHMS
- FREQUENCY RANGE: 200 TO 4000 Hz

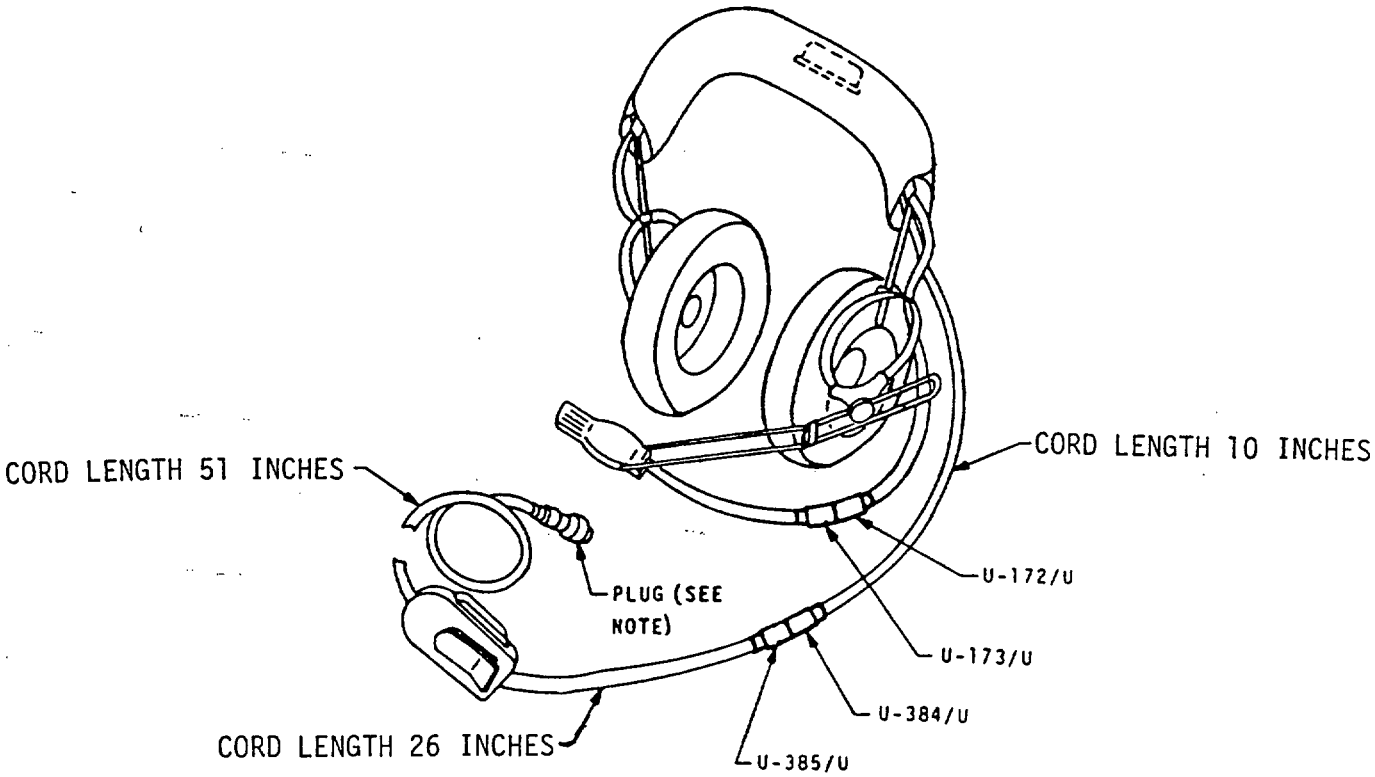
CONTROLLING DOCUMENT: REB-2688034

NOTES:
MATES WITH JACKS PER MIL-J-641 MATING TABLE.

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HEADSET-MICROPHONE

H-172A/AIC



PART NUMBER: H-172A/AIC

DESCRIPTION: HEADSET-MICROPHONE, DYNAMIC, NOISE CANCELLING WITH TRANSISTORIZED AMPLIFIER; LOW DISTORTION. EARPHONE IN RIGHT EARPIECE ONLY; LEFT EARPIECE IS PERFORATED FOR AMBIENT LISTENING. DPST MICROPHONE SWITCH HAS CLIP TO ATTACH TO CLOTHING.

EARPHONE: (ONE ONLY)
IMPEDENCE: 600 OHMS
FREQUENCY RANGE: 100 TO 4500 Hz

MICROPHONE:
FREQUENCY LOAD: 30 OHMS
FREQUENCY RANGE: 300 TO 4500 Hz
IM DISTORTION: 2% MAX

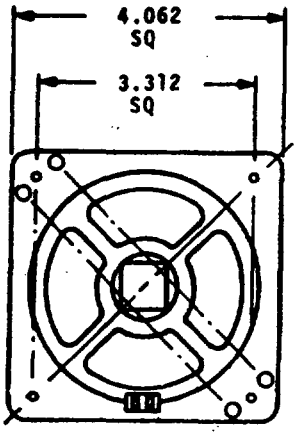
CONTROLLING DOCUMENT: MIL-H-83511/5

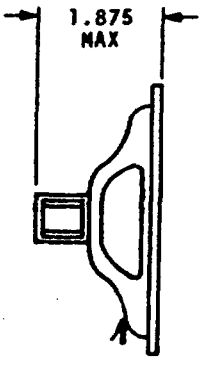
NOTE:
MATES WITH MS3100E14S-5P OR EQUIVALENT.

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LOUDSPEAKER

LS-455/U





PART NUMBER: LS-455/U

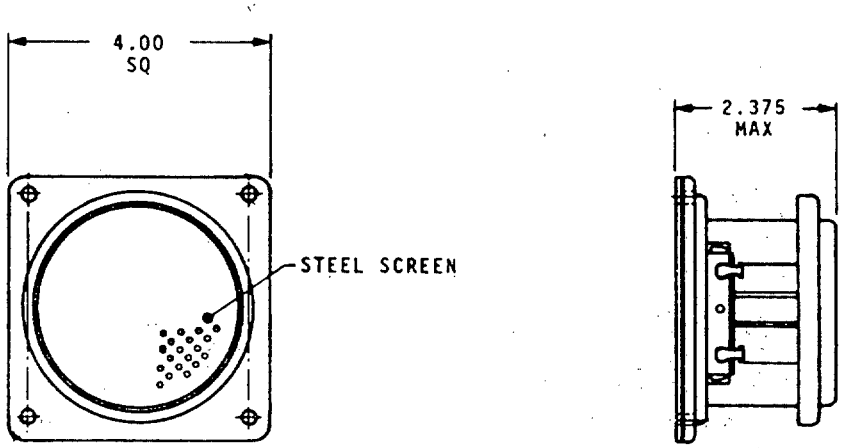
DESCRIPTION: LOUDSPEAKER, PERMANENT MAGNET, 4 INCH
VOICE COIL:
IMPEDANCE: 3.2 OHMS
FREQUENCY RANGE: 200 TO 5000 Hz
POWER: 2 WATTS AVERAGE, 5 WATTS PEAK

CONTROLLING DOCUMENT: RE-C2687531

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LOUDSPEAKER

LS-445/U



PART NUMBER: LS-445/U

DESCRIPTION: LOUDSPEAKER, PERMANENT MAGNET, 4 INCH, FUNGUS- GUNBLAST -, AND IMMERSION-RESISTANT.

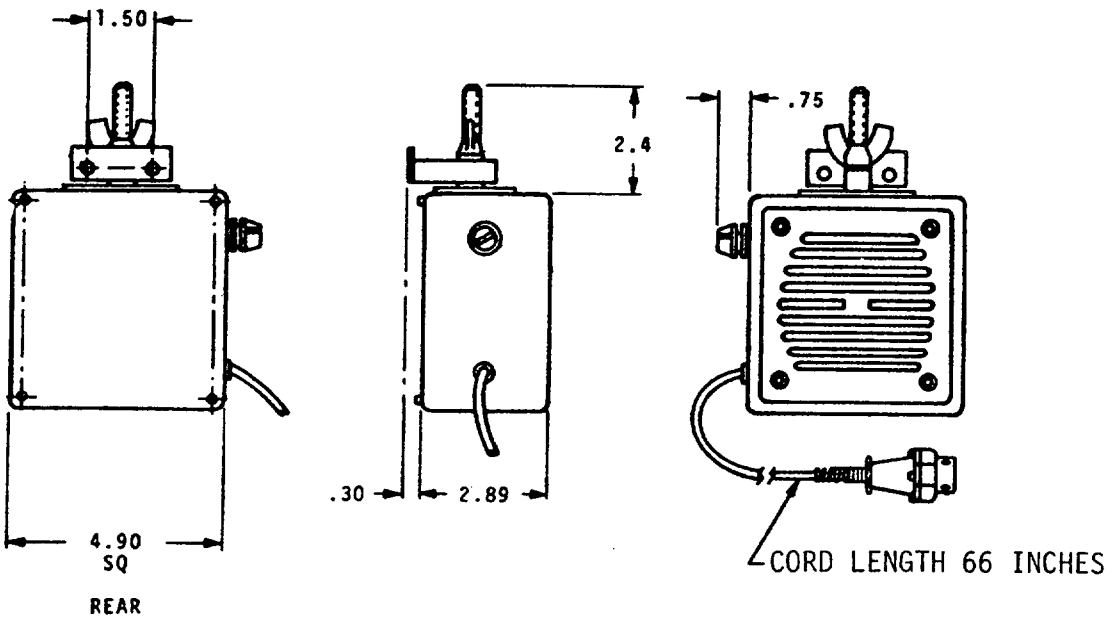
VOICE COIL:
IMPEDANCE: 8 OHMS
FREQUENCY RANGE: 300 TO 7000 Hz
POWER: 2 WATTS AVERAGE, 4 WATTS PEAK

CONTROLLING DOCUMENT: RE-C2687531 AND MIL-L-12606D-1

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LOUDSPEAKER

LS-166/U AND LS-454/U



PART NUMBER: LS-166/U AND LS-454/U

DESCRIPTION: LOUDSPEAKER, PERMANENT MAGNET, 4 INCH, FUNGUS-, GUNBLAST-, AND IMMERSION-RESISTANT.

INPUT:

IMPEDENCE: 600 OHMS
FREQUENCY RANGE: 300 TO 7000 Hz
POWER: 2 WATTS

CONTROLLING DOCUMENT: MIL-L-12632/2 AND /4 CONTAIN LS-445/U LOUDSPEAKER UNIT PER MIL-L-12606 MATCHING TRANSFORMER AND SPDT INPUT SELECTOR SWITCH. (SEE TABLE I.)

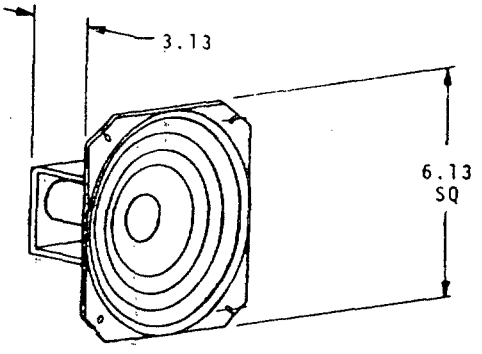
TABLE I. INPUT SELECTOR SWITCH.

TYPE NO.	PLUG	INPUT SELECTOR SWITCH
LS-166/U	U-77/U	REQUIRED
LS-454/U	U-229/U	NOT REQUIRED

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LOUDSPEAKER

NT-491814



PART NUMBER: NT-491814

DESCRIPTION: LOUDSPEAKER, PERMANENT MAGNET, 6 INCH
VOICE COIL:
IMPEDANCE: 3.2 OHMS
FREQUENCY RANGE: 200 TO 5000 Hz
POWER: 4 WATTS AVERAGE
CONTROLLING DOCUMENT: RE-C2687571
PROCUREMENT SPECIFICASTION: MIL-L-17061

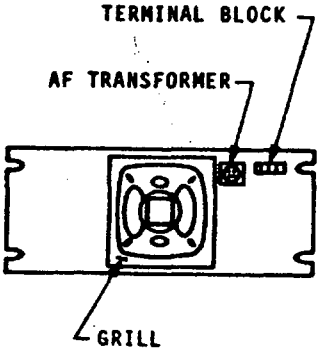
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LOUDSPEAKER

LS-169/U



FRONT VIEW



REAR VIEW

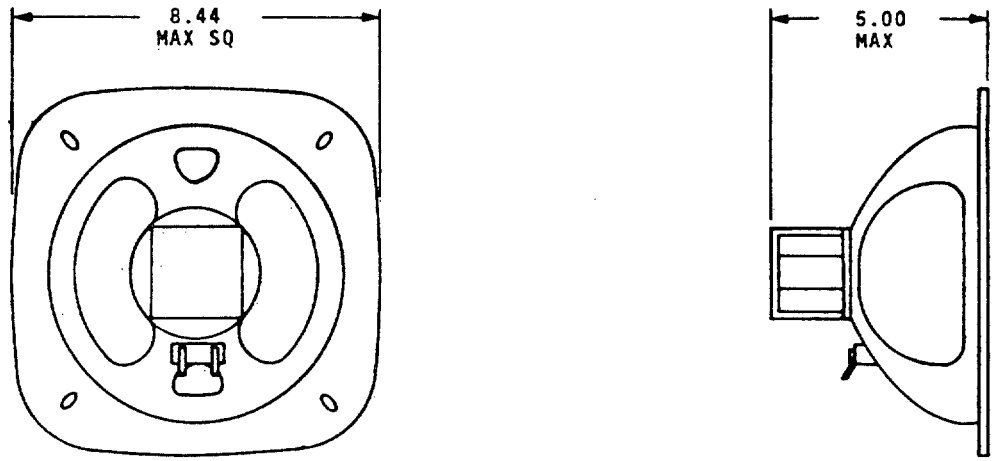
PART NUMBER: LS-169/U

DESCRIPTION: LOUDSPEAKER, PANEL MOUNTED, PERMANENT MAGNET, 6-INCH
INPUT:
IMPEDENCE: 600 OHMS
FREQUENCY RANGE: 200 TO 5000 Hz
POWER: 4 WATTS AVERAGE
CONTROLLING DOCUMENT: REF-268780. CONTAINS NT-49181A LOUDSPEAKERS AND MATCHING TRANSFORMER.
NOTES:
1. PANEL: 19 X 6.5 INCHES, ALUMINUM
2. MOUNTING SLOTS: .438 X .25 LOCATED AT CORNES OF AN 18.25 X 4 INCH RECTANGLE.
3. TOTAL DEPTH: 4.375 INCHES

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LOUDSPEAKER

LS-180/U



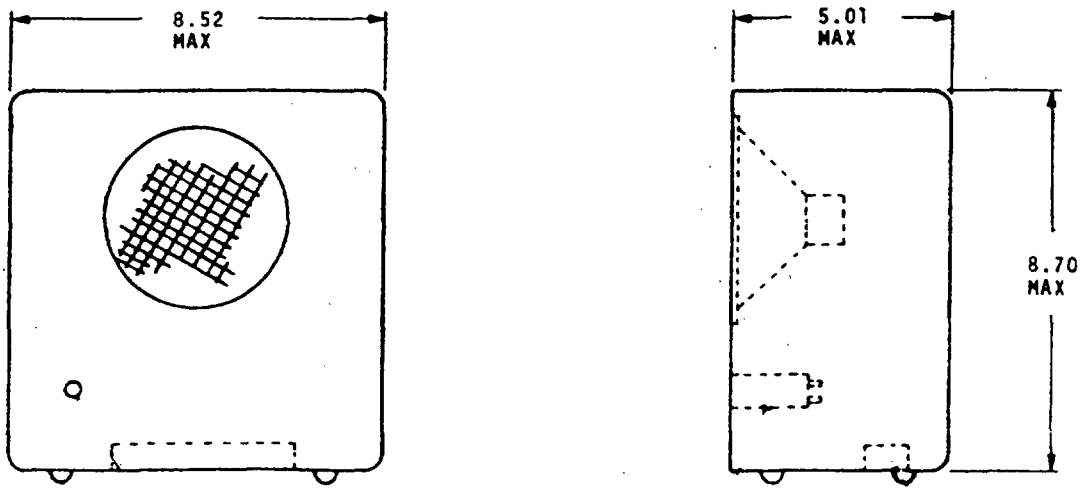
PART NUMBER: LS-180/U

DESCRIPTION: LOUDSPEAKER, PERMANENT MAGNET, 8-INCH
VOICE COIL:
IMPEDENCE: 8 OHMS
FREQUENCY RANGE: 150 TO 7000 Hz
POWER: 5 WATTS PEAK - 10 WATTS
CONTROLLING DOCUMENT: MIL-L-10911/3

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LOUDSPEAKER

LS-474/U



PART NUMBER: LS-474/U

DESCRIPTION: LOUDSPEAKER, ENCLOSED, PERMANENT MAGNET, 6 INCH; SUPPLIED WITH MOUNTING BRACKETS FOR WALL, RACK PANEL OR TABLE TOP MOUNTING. TWO UNITS SIDE BY SIDE MAY BE MOUNTED ON CY-597/G RELAY BACK. PROVISION IS MADE TO ADD VOLUME CONTROL BY REMOVING HOLE PLUG.

INPUT:

IMPEDENCE: 600 OHMS OR 3.2 OHMS (SELECTED BY JUMPERS)
FREQUENCY RANGE: 200 TO 5000 Hz
POWER: 4 WATTS AVERAGE

CONTROLLING DOCUMENT: REF 2687574

REF INFORMATION: CONTAINS NT-491814 LOUDSPEAKER, 3.2 OHMS, AND MATCHING, 600 OHMS, TRANSFORMER.

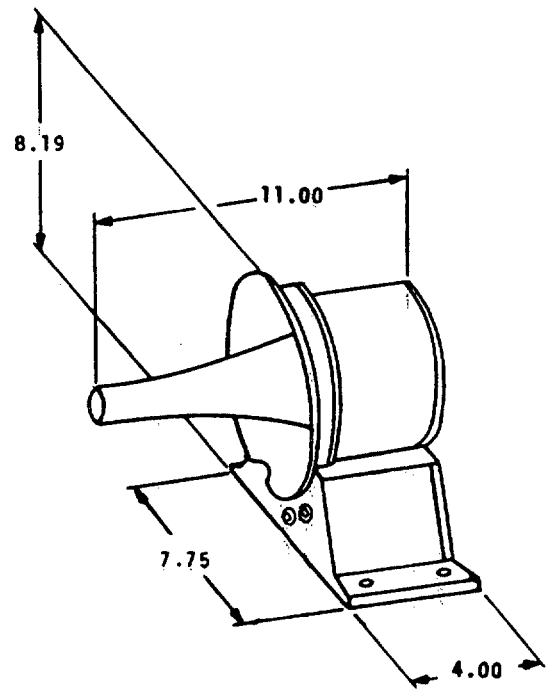
VOLUME CONTROL: MALLORY TYPE L4 PAD OR EQUIVALENT, AND KNOB MS91528-1N26 (MUST BE ORDERED SEPARATELY, IF REQUIRED).

NOTE: WITH EACH UNIT, TWO MOUNTING BRACKETS ARE SUPPLIED WHICH MAY BE EXTERNALLY ATTACHED TO THE FRONT OR REAR OF THE UNIT BY DRILLING HOLES IN THE SIDES. TWO UNITS MAY BE SCREWED TOGETHER TO FORM A DOUBLE UNIT FOR RACK MOUNTING. FULL INSTRUCTIONS ARE INCLUDED WITH EACH UNIT.

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LOUDSPEAKER

NT-49546



PART NUMBER: NT-49546

DESCRIPTION: LOUDSPEAKER, REFLEX HORN WITH RADIAL DEFLECTOR TO GIVE 360 DEGREES HORIZONTAL DISPERSION; FOR USE ONLY IN LOCATIONS EXPOSED TO GUNBLAST AND WEATHER.

VOICE COIL:
IMPEDENCE: 16 OHMS
FREQUENCY RANGE: 200 TO 5000 Hz
POWER: 10 WATTS

CONTROLLING DOCUMENT: MIL-L-16225

SUPPLIED COMPLETE WITH VOLUME CONTROL AND MATCHING TRANSFORMER WITH PRIMARY TAPPED FOR 600 OHMS/1,200 OHMS/ 1,800 OHMS/2,400 OHMS/3,000 OHMS INPUT IMPEDANCE.

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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